Highlights

Refreshed Søreide-Whitson Framework for Gas Solubility in Water and Brine with Extension to Hydrogen

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- Drop-in BIP correlations fitted on the original Søreide-Whitson procedure
- Framework extended to H_2 for underground hydrogen storage applications
- H_2S salting-out support added, absent in original Søreide-Whitson framework
- Embedded $\Delta k_{ij}(T, m)$ for brine reproduces Sechenov models within 3%
- BIP absorbs upstream model differences, validating drop-in approach

Refreshed Søreide-Whitson Framework for Gas Solubility in Water and Brine with Extension to Hydrogen

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Abstract

Refit draft. This version refits the published correlations (Fluid Phase Equilib. 114824) on the coupled equilibrium, the simultaneous application of the two S&W BIP sets described by Whitson and Brulé (Section 3, which also discusses the alternatives): one equilibrium with k_{ij}^{AQ} in the aqueous phase and k_{ij}^{NA} in the non-aqueous phase, the vapour carrying its equilibrium water. The published article regressed k_{ij}^{AQ} against a dry pure-gas vapour. Data, exclusions, functional forms and fitting routines are unchanged. Appendix Appendix C lists every change.

The Søreide-Whitson (S&W, 1992) equation of state framework for gas-water-brine phase equilibria is widely used in reservoir simulation; published updates since have addressed individual gases rather than the framework as a whole. This work presents updated $k_{ij}^{\mathrm{AQ}}(T)$ and embedded salinity $\Delta k_{ij}(T, m)$ correlations for eight gases (CH_4, C_2H_6, C_3H_8, n-C_4H_{10}, CO_2, H_2S, N_2, H_2), developed from approximately 2,000 pointwise-regressed BIP values. Helium is treated on the same basis in an appendix. The correlations retain the original S&W water alpha function and the coupled equilibrium procedure, so they can replace the original coefficients in existing simulator implementations; only the BIP correlation coefficients need updating.

Freshwater k_{ij}^{AQ} correlations are fitted to freshwater data only, and salinity effects are represented through an additive $\Delta k_{ij}(T, m)$ term fitted to reproduce modern Sechenov models for each gas. Against freshwater solubility data, scored on the same procedure, the new correlations give MARE of 12.4% for CO_2 (n = 606; S&W 15.3%), 6.3% for H_2S above x = 0.003 (n = 385; S&W 7.8%), 3.9% for CH_4 (S&W 3.9%), 5.3% for N_2 (7.2%), 8.4% for C_2H_6 (11.3%) and 3.4% for C_3H_8 (8.9%). Replacing the S&W water alpha function with a higher-accuracy form and adopting a modular gamma-phi salting-out framework did not improve solubility predictions once the BIP was refitted, which supports the simpler embedded approach.

The framework is extended to hydrogen (MARE 11.3%, n = 142) for underground storage applications, and H_2S salting-out support is added for the first time.

Keywords: gas solubility, Søreide-Whitson, binary interaction parameter, Sechenov coefficient, equation of state, underground hydrogen storage, reservoir simulation

1. Introduction

The Søreide-Whitson (S&W) framework [1] provides a practical approach for calculating gas-water-brine phase equilibria using the Peng-Robinson equation of state (PR-EOS) with modified parameters for water and temperature-dependent binary interaction parameters (BIPs). Since its publication in 1992, this framework has been widely adopted in commercial reservoir simulators for applications including CO_2 sequestration, acid gas injection, and natural gas processing. Published updates have since addressed individual gases (e.g., CO_2 [19]; H_2 [13, 18]) rather than the framework as a whole.

Three decades of accumulated experimental data and evolving application requirements motivate a reassessment of the S&W framework:

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1. **Expanded experimental database.** The original S&W correlations were fitted to datasets of a few dozen points per gas. Three decades of additional experimental data have become available since; this work compiles approximately 2,000 pointwise-regressed BIP values across eight gases.
2. **Salting-out model evolution.** The original S&W framework already provided gas-specific salting-out via Eq. 13 for hydrocarbons and embedded salinity terms in the BIP correlations for CO_2 (Eq. 8) and N_2 (Eq. 7). However, their embedding ties the salinity correction to the freshwater k_{ij} functional form, and H_2S (Eq. 9) had no salinity support. Modern Pitzer-based Sechenov models are now available for CO_2 and H_2S. This work provides updated embedded $\Delta k_{ij}(T, m)$ correlations that reproduce these modern models while remaining drop-in compatible with the original framework.
3. **Hydrogen support.** Underground hydrogen storage (UHS) in depleted reservoirs and saline aquifers is an emerging application requiring validated H_2-water-brine parameters not included in the original framework. Helium, added after the published version, is treated on the same basis in Appendix Appendix B.

All correlations are fitted and scored with the coupled equilibrium, in which each phase carries its own S&W BIP set (the equilibrium NeqSim and pyResToolbox solve), so the new coefficients replace the 1992 coefficients like for like (Section 3).

2. Theoretical Background

2.1. Peng-Robinson Equation of State

The Peng-Robinson equation of state [2] is given by:

$$P = \frac{RT}{V - b} - \frac{a(T)}{V(V + b) + b(V - b)} \tag{1}$$

where P is pressure, T is temperature, V is molar volume, R is the universal gas constant, and $a(T)$ and b are the equation of state parameters.

For pure components:

$$a_i = 0.45724 \frac{R^2 T_{c,i}^2}{P_{c,i}} \alpha_i(T) \tag{2}$$

$$b_i = 0.07780 \frac{R T_{c,i}}{P_{c,i}} \tag{3}$$

where $T_{c,i}$ and $P_{c,i}$ are the critical temperature and pressure of component i, and $\alpha_i(T)$ is the temperature-dependent alpha function.

2.2. Water Alpha Function

The alpha function for water controls the PR-EOS vapour pressure prediction and thus the fugacity of water in both liquid and vapour phases; its accuracy affects every dissolved-gas and water-content prediction.

2.2.1. Original S&W Formulation

S&W [1] proposed a modified alpha function that embeds both temperature and salinity dependence in a single expression:

$$\sqrt{\alpha_{H_2O}} = 1 + 0.4530 \left[1 - T_r \left(1 - 0.0103\, c_{sw}^{1.1}\right)\right] + 0.0034 \left(T_r^{-3} - 1\right) \tag{4}$$

where $T_r = T/T_{c,H_2O}$ is the reduced temperature and c_{sw} is the salt molality (mol/kg H_2O). The water critical properties used throughout this work follow S&W Table 3: $T_{c,H_2O} = 647.3$ K, $P_{c,H_2O} = 221.2$ bar (22.12 MPa), $\omega_{H_2O} = 0.3434$.

The functional form employs a T_r^{-3} inversion term rather than the standard Soave $(1 - \sqrt{T_r})$ basis, which prevents adopting improved alpha functions developed for polar molecules without refitting the entire correlation. The salinity correction $(0.0103\, c_{sw}^{1.1})$ is embedded directly in the temperature-dependent term. S&W fitted the salinity coefficients to the brine steam tables of Haas [31]. The salinity coefficient and exponent can be retuned independently of the freshwater parameters, but the temperature dependence of the brine correction is locked to T_r by the equation structure.

2.2.2. Deficiencies

Comparison of the S&W freshwater alpha ($c_{sw} = 0$) against the IAPWS-95 [29] reference standard for pure water vapour pressure reveals a mean absolute relative error (MARE) of 1.18% over 10–340°C, with errors exceeding 5% at low temperatures. While adequate for reservoir engineering applications, this accuracy is a limiting factor when the alpha function is used as a baseline for independent salinity corrections. Additionally, the Katz correlation [32] used by S&W for brine water activity ($a_w \approx 1 - 0.02865\, c_{sw}^{1.44}$, their Eq. 16) is temperature-independent and becomes unreliable above approximately 10 wt% NaCl.

2.2.3. Investigation: Effect of Alpha Function on Solubility

To test whether improved water vapour pressure accuracy improves gas solubility predictions, we fitted a Mathias-Copeman [30] three-parameter alpha function to IAPWS-95 [29] reference vapour pressures, reducing the freshwater MARE (evaluated over 20–200°C) from 1.35% to 0.006%. We also tested a modular gamma-phi salting-out framework using gas-specific Sechenov models in place of the embedded BIP approach. However, when the k_{ij}^{AQ} correlations are refitted, solubility predictions are indistinguishable regardless of which alpha function or salting-out implementation is used (Table 1).

Table 1: Effect of water alpha function on solubility MARE (%). *This investigation was run with the dry-vapour solve of the published article and was not repeated on the coupled procedure; the numbers are those of the published version and are not comparable with the other tables of this draft.* "S&W alpha" uses the original S&W alpha (Eq. 4) with refitted k_{ij} correlations (this work). "MC-3 alpha" uses the Mathias-Copeman three-parameter alpha with independently refitted k_{ij}. Brine predictions use the embedded Δk_{ij} correction (Section 7) for the S&W-alpha columns and the explicit Sechenov correction for the MC-3 columns.

	Freshwater MARE (%)		Brine MARE (%)	
Gas	S&W alpha	MC-3 alpha	S&W alpha	MC-3 alpha
CO_2	16.4	16.4	17.3	17.2
H_2S	38.5	38.7	15.4	14.6
CH_4	4.8	5.4	5.6	5.1
N_2	5.7	5.9	4.6	5.6
H_2	10.2	10.2	4.1	5.2
C_2H_6	12.9	12.8	—	—
C_3H_8	5.0	4.7	—	—

The regressed k_{ij} compensates for systematic bias in the water fugacity, so the vapour pressure improvement does not carry through to solubility. The original S&W alpha function and the embedded BIP approach are therefore retained for all correlations in this work, keeping them drop-in compatible with existing S&W implementations.

2.2.4. Brine Vapour Pressure Depression

The S&W alpha function (Eq. 4) includes an embedded salinity term ($0.0103\, c_{sw}^{1.1}$) that accounts for brine vapour pressure depression. This is retained in the present work.

2.3. Binary Interaction Parameters

For mixtures, the attractive parameter uses the classical mixing rule:

$$a = \sum_i \sum_j x_i x_j (a_i a_j)^{0.5} (1 - k_{ij}) \tag{5}$$

where x_i and x_j are mole fractions and k_{ij} is the binary interaction parameter.

The S&W framework uses separate BIPs for aqueous (k_{ij}^{AQ}) and non-aqueous (k_{ij}^{NA}) phases, recognising that gas-water interactions differ fundamentally between the two phases. The original S&W [1] correlations parameterised k_{ij}^{AQ} as functions of reduced temperature $T_{ri} = T/T_{c,i}$ with embedded salinity terms. For hydrocarbons (C_1–nC_4):

$$k_{ij}^{AQ} = A_0(1 + \alpha_0\, c_{sw}) + A_1\, T_{ri}(1 + \alpha_1\, c_{sw}) + A_2\, T_{ri}^2(1 + \alpha_2\, c_{sw}) \tag{6}$$

where $A_0 = 1.1120 - 1.7369\,\omega_i^{-0.1}$, $A_1 = 1.1001 + 0.8360\,\omega_i$, $A_2 = -0.15742 - 1.0988\,\omega_i$, $\alpha_0 = 0.017407$, $\alpha_1 = 0.033516$, $\alpha_2 = 0.011478$ (errata-corrected values), ω_i is the acentric factor, and c_{sw} is the salt molality. For $c_{\mathrm{sw}} = 0$ this reduces to a freshwater quadratic in T_{ri}.

For N_2:

$$k_{ij}^{\mathrm{AQ}} = -1.70235\,(1 + 0.025587\,c_{\mathrm{sw}}^{0.75}) + 0.44338\,(1 + 0.08126\,c_{\mathrm{sw}}^{0.75})\,T_{ri} \tag{7}$$

For CO_2:

$$k_{ij}^{\mathrm{AQ}} = -0.31092\,(1 + 0.15587\,c_{\mathrm{sw}}^{0.7505}) + 0.23580\,(1 + 0.17837\,c_{\mathrm{sw}}^{0.979})\,T_{ri} - 21.2566\,\exp(-6.7222\,T_{ri} - c_{\mathrm{sw}}) \tag{8}$$

For H_2S (freshwater only; no salinity term):

$$k_{ij}^{\mathrm{AQ}} = -0.20441 + 0.23426\,T_{ri} \tag{9}$$

These correlations embed salinity directly into the BIP for all gases except H_2S. Hydrogen was not included in the original framework.

Table 2 lists the component properties used throughout this work. Because the fitted BIP correlations absorb any bias in the pure-component parameters, they are valid only in combination with the tabulated values.

Table 2: Component properties used in this work. T_c, P_c, ω and T_b follow S&W Table 3 [1]; H_2 (absent from S&W) uses NIST values. T_b enters only the Sechenov correlation (Eq. 13).

Component	T_c (K)	P_c (bar)	ω	T_b (K)
CO_2	304.2	73.8	0.2273	194.7
H_2S	373.2	89.4	0.1081	212.8
N_2	126.1	34.0	0.0403	77.3
CH_4	190.6	46.0	0.0108	111.6
H_2	33.145	12.964	−0.219	20.3
C_2H_6	305.4	48.8	0.0998	184.6
C_3H_8	369.8	42.5	0.1517	231.1
nC_4H_{10}	425.2	38.0	0.1931	272.7
H_2O	647.3	221.2	0.3434	—

2.4. Equilibrium Procedure

S&W [1] give each phase its own mixture parameter: a^{NA} from the non-aqueous composition with k_{ij}^{NA} and a^{AQ} from the aqueous composition with k_{ij}^{AQ} (their Eqs. 10a and 10b). Applying both in one equilibrium, as Whitson and Brulé [3] describe, gives

$$x_i\,\phi_i^L\left(T, P, \mathbf{x};\, k^{\mathrm{AQ}}\right) = y_i\,\phi_i^V\left(T, P, \mathbf{y};\, k^{\mathrm{NA}}\right) \tag{10}$$

for every component, gas and water. The non-aqueous phase carries its equilibrium water, so the gas fugacity it imposes on the aqueous phase is that of the water-saturated gas. Both BIP sets act in one calculation; this is called the coupled procedure below. Every correlation in this work is regressed and scored with Eq. 10. Other procedures are in use, and the 1992 paper does not record which one the original regression used; Section 3 sets out the alternatives, the evidence and the reasons for this choice.

2.5. Sechenov (Salting-Out) Equation

Gas solubility in brine is reduced relative to pure water following the Sechenov relationship:

$$\log_{10}\left(\frac{x_{\mathrm{fresh}}}{x_{\mathrm{brine}}}\right) = k_s \cdot m \tag{11}$$

where x_{fresh} and x_{brine} are gas mole fractions in freshwater and brine respectively, k_s is the Sechenov coefficient (kg/mol), and m is salt molality (mol/kg H_2O).

S&W [1] provided a generalised correlation for k_s that depends only on the normal boiling point of the gas. The original correlation is given in Fahrenheit units:

$$k_s = 0.13163 + 4.45 \times 10^{-4} T_b - 7.692 \times 10^{-4} T_F + 2.6614 \times 10^{-6} T_F^2 - 2.612 \times 10^{-9} T_F^3 \tag{12}$$

where T_F is temperature in °F and T_b is the normal boiling point in K. Converting to SI units ($T_F = 1.8\,T - 459.67$), this becomes:

$$k_s = 1.3012 + 4.45 \times 10^{-4} T_b - 8.769 \times 10^{-3} T + 2.0293 \times 10^{-5} T^2 - 1.5233 \times 10^{-8} T^3 \tag{13}$$

where T is temperature in K. This formulation treats gas-specific effects through the vertical offset (T_b term) whilst assuming a common temperature dependence for all gases, an approximation adequate for the non-polar gases (hydrocarbons, N_2, H_2) but not for the acid gases CO_2 and H_2S (Sections 6.2 and 6.3).

3. Choice of Equilibrium Procedure

The correlations are intended to replace the 1992 coefficients like for like, so they must be fitted on the equilibrium that an S&W implementation solves. Three procedures appear in the literature, in practice and in the two versions of this work (Table 3). They share the EOS, the water alpha and the BIP correlations and differ only in how the two BIP sets enter the phase-equilibrium calculation.

Table 3: Equilibrium procedures for the S&W model.

Procedure	Calculation	Where it appears
Coupled (Eq. 10)	One equilibrium; k^{AQ} in the aqueous phase, k^{NA} in the non-aqueous phase	Whitson and Brulé [3] (simultaneous application); NeqSim [43]; pyResToolbox 3.10 [44]; this work
Two-flash	Flash A with k^{AQ} in both phases gives the aqueous composition; flash B with k^{NA} in both phases gives the non-aqueous composition	Whitson and Brulé [3] (fallback for a single BIP set); EOS simulators that carry one BIP set per pair [42]
Dry vapour	Aqueous phase with k^{AQ} against pure, dry gas at total pressure	Published version of this work [34]

3.1. What Is Known About the 1992 Regression

The 1992 paper defines the phase-specific mixture parameters (Eqs. 10a/10b) and the BIP correlations, but does not state how the phase equilibrium was solved, either in application or when k_{ij}^{AQ} was regressed on solubility data. Eqs. 10a/10b hold for the coupled form and for the phase each flash of the two-flash form retains, so they do not separate the two. S&W adopt the two-BIP-set approach of Peng and Robinson and note that two values of a may be thermodynamically inconsistent, an issue that arises only when both meet in one equilibrium; this points to the coupled form but does not state it. Whitson and Brulé [3] are explicit: simultaneous application of the aqueous and non-aqueous BIPs requires modifying a standard single-BIP-set EOS implementation, their mutual-solubility figures use it, and a standard implementation can instead use k_{ij}^{AQ} in both phases when only gas solubility is needed (or k_{ij}^{NA} when only water content is needed), at the cost of an inaccurate composition for the other phase. The second S&W author describes the two-flash form as the way the model is applied to reservoir mixtures in practice, because most commercial EOS simulators carry a single BIP set per pair, and recalls the original regression as two separate flashes [42].

The coefficients themselves were tested for which procedure they are consistent with. The S&W 1992 k_{ij}^{AQ}, with the S&W k_{ij}^{NA}, were scored on the freshwater evaluation set with the coupled procedure and with flash A of the two-flash form (Table 4). For CO_2, which has its own correlation form, the coupled procedure fits better (MARE 15.3%

against 18.4%), and the difference sits at high pressure: above 400 bar the bias is −3.0% coupled and −14.1% two-flash (n = 178). Pre-1992 sources show the same (Takenouchi and Kennedy 1964, +4.0% against −7.6%; Wiebe and Gaddy 1940, −5.3% against −10.8%). C_2H_6 follows CO_2 (Culberson 1950 above 200 bar, −2.5% against −8.5%). C_3H_8 goes the other way (Kobayashi 1953, +4.8% against −0.5%), and so, weakly, does CH_4 above 200 bar. H_2S and N_2 do not distinguish the two. The hydrocarbon k_{ij}^{AQ} is one generalised correlation in ω fitted to all hydrocarbons together, so it is a weaker test than CO_2. The scoring uses today's data set, which is larger than the 1992 one. The test therefore does not identify the procedure used in 1992; it shows the 1992 coefficients are at least as consistent with the coupled procedure as with the two-flash form.

Table 4: Freshwater solubility MARE (%), bias in brackets, with the coupled procedure (Eq. 10) and the two-flash form (flash A for the aqueous composition). S&W 1992 coefficients with S&W k_{ij}^{NA}; this work with Table 8.

		S&W 1992		This work	
Gas	n	Coupled	Two-flash	Coupled	Two-flash
CO_2	611	15.3 (−6.3)	18.4 (−10.3)	12.4 (+4.8)	13.0 (+0.2)
H_2S	405	30.8 (+20.6)	31.1 (+19.9)	31.9 (+25.7)	32.1 (+24.7)
CH_4	115	3.9 (−0.6)	3.9 (−2.0)	3.9 (−0.7)	4.1 (−2.1)
N_2	127	7.2 (+3.7)	7.1 (+3.0)	5.3 (+1.2)	5.2 (+0.4)
C_2H_6	95	11.3 (−5.5)	14.0 (−9.2)	8.4 (+0.8)	9.9 (−3.1)
C_3H_8	59	8.9 (+4.8)	7.4 (−0.5)	3.4 (+1.2)	7.1 (−3.8)

3.2. Available Implementations

NeqSim [43] solves one flash and chooses the BIP set per phase inside its S&W mixing rule: the aqueous correlation for a phase whose water mole fraction exceeds 0.8, the non-aqueous value otherwise, with the S&W water alpha (at the aqueous salinity) in every phase. pyResToolbox 3.10 [44] also solves one flash, assigning k_{ij}^{AQ} to a liquid-root phase in which water is the majority component and k_{ij}^{NA} to every other phase. Both are the coupled procedure; they differ from Eq. 10 as solved here only in how a phase is labelled aqueous. On the freshwater set the NeqSim rule changes the label on 3 of 1,684 points, where the vapour holds more than 80% water (CO_2, Truche 2015, 200°C and 17 bar; H_2S, Lee 1977, 180°C, and Zezin 2011, 200°C), and moves those predictions by at most 1.0 percentage point. Root selection differs on the three-phase line (Section 8). Reaktoro [45] applies one constant set, the S&W non-aqueous values, to every phase, and so does not implement the S&W aqueous model. NeqSim offers the coefficients of the published version of this work as an option; those were fitted on the dry-vapour basis and are biased in its coupled flash (Appendix Appendix C), and the coefficients of this draft supersede them for such use.

3.3. Procedure Adopted

This work standardises on the coupled procedure, for four reasons. It is the simultaneous application that Whitson and Brulé describe as the method, with the two-flash form as the fallback for a single-BIP-set implementation. It is the only one of the three that returns both phase compositions from one equilibrium closing one mass balance. It is what NeqSim and pyResToolbox solve, so the coefficients drop into those implementations unchanged. And on the evidence of Table 4 the 1992 coefficients are at least as consistent with it as with the two-flash form, so scoring them on it does not penalise them.

A simulator restricted to one BIP set per pair can use the two-flash form with these coefficients. The cost is the two-flash columns of Table 4: freshwater MARE rises by 0.6 percentage points for CO_2, 1.5 for C_2H_6 and 3.7 for C_3H_8, with no material change for H_2S, CH_4 and N_2. Even so, the refit coefficients score at or below the 1992 coefficients in the two-flash form for every gas except H_2S (32.1% against 31.1%) and CH_4 (4.1% against 3.9%). The difference between the two forms is set by the effect of the BIP on the fugacities in the non-aqueous phase, not by its water content alone: for CO_2 it grows with pressure into the dense-phase region, where the gas holds the least water.

4. Methodology

All correlation fits, validation metrics, and figures in this work are restricted to temperatures at or below 200°C (473 K). This upper bound corresponds to the practical limit for most subsurface gas storage and production applications. CO_2 (T_c = 304 K) and H_2S (T_c = 373 K) have critical points within this range, and the PR-EOS exhibits structural accuracy limitations near and above these temperatures, particularly at high gas partial pressures where vapour-liquid-liquid equilibrium may occur (Section 8.7).

Three data sources are excluded: Prutton & Savage (1945) [26] for CO_2 (measurements in $CaCl_2$ solutions rather than NaCl; on an NaCl-molality basis the implied salting-out is anomalously strong and dominates the brine MARE), Blount & Price (1982) [24] and McGee et al. (1981) [25] for CH_4 (systematic bias), and Barrett (1988) [27] for H_2S (all 199 brine measurements at 1 bar, where the cubic EOS has structural limitations for dissolved-gas predictions at near-atmospheric partial pressures). Five CO_2 points of Wang et al. (2021) at 160–200°C and 5–10 bar are also excluded: the stated pressure is at or below the water vapour pressure (6.2 bar at 160°C, 15.5 bar at 200°C), so it must be a CO_2 partial pressure, and no aqueous phase exists at that total pressure.

4.1. Pointwise BIP Regression

For each experimental data point (temperature T, pressure P, measured solubility x_{exp}), we regressed the k_{ij}^{AQ} for which Eq. 10 reproduces x_{exp}, with k_{ij}^{NA} at its recommended value (Section 5.11):

$$f_i^L(T, P, \mathbf{x}; k_{ij}^{\text{AQ}}) = f_i^V(T, P, \mathbf{y}; k_{ij}^{\text{NA}}), \quad i = \text{gas, water} \tag{14}$$

where f_i^L and f_i^V are fugacities in the aqueous and non-aqueous phases. The equilibrium is solved by successive substitution on $K_i = \phi_i^L/\phi_i^V$ with the binary Rachford-Rice solution, falling back to a nested bracketed solve; the outer regression brackets k_{ij}^{AQ} on a grid and refines with Brent's method. Near the water-saturated mixture critical locus (nine H_2S points of Selleck 1952 and Koschel 2013, 104–138°C, 155–300 bar) the root is located by bisection towards the edge of the solution region. All points in the fit set converge. The solver reproduces the seven water contents and solubility ratios of an independent implementation [35] to the printed digits, with fugacity residuals below 2×10^{-13} in $\ln f$. The non-aqueous phase takes the vapour root of the cubic. Because k_{ij}^{NA} was not changed by the re-check of Section 5.11, no further iteration between the two BIP sets was needed.

This pointwise regression yields a database of (T, k_{ij}) pairs that directly reflects the experimental data without imposing a correlation form. Subsequent correlation fitting operates on these regressed values rather than on raw solubility data, separating the EOS inversion step from the functional form selection.

4.2. Deconvolution of Freshwater BIP and Salting-Out

The freshwater BIP (k_{ij}^{AQ}) is explicitly separated from salting-out effects (k_s). In the original S&W framework, salinity dependence was embedded directly in the BIP correlations (Eqs. 6–8), implemented in a way that ties the salinity correction to the freshwater k_{ij} functional form. Regardless of whether the original authors fitted these components sequentially or simultaneously, this embedding means the implied Sechenov coefficients, while of the correct order of magnitude, do not reproduce the temperature-dependent character of the salting-out behaviour established by modern Pitzer models or seen in experimental data.

Our approach:

1. Regress k_{ij}^{AQ} using freshwater data only, establishing the gas-water BIP as a function of temperature alone.
2. Independently validate or update the Sechenov coefficient k_s using modern Pitzer models as benchmarks.
3. Apply the salinity correction as a separate layer.

The freshwater BIP and the salting-out model are then independent: updating one does not require re-fitting the other.

4.3. Correlation Fitting

Correlation coefficients were determined by L1 (MAE-minimising) regression rather than L2 (least-squares). L1 regression is more robust to outliers in the pointwise-regressed k_{ij} values, which may include experimental artefacts or conditions where the EOS model is inadequate. The error metrics used throughout this work are:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \qquad \text{MARE} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \% \tag{15}$$

where y_i is the measured value and $\hat{y}_i$ is the predicted value.

For a correlation of the form $k_{ij} = g(T;\theta)$ with parameters θ, the objective function is:

$$\min_{\theta}\sum_{i=1}^{n}\left|k_{ij,\text{regressed},i} - g(T_i;\theta)\right| \tag{16}$$

4.4. Experimental Data Sources

Table 5 summarises the experimental database compiled for this work.

Three point-count bases are used throughout: (i) the full converged database, every point for which the pointwise k_{ij}^{AQ} regression converged, at any temperature (Table 5); (ii) the fit set, the subset at $T \leq 200°\text{C}$ used to fit and score the correlations (Tables 6 and A.13); and (iii) for salting-out, the matched freshwater/brine pairs retained after the source exclusions above. Each table caption states the basis used.

Table 5: Summary of experimental data used for pointwise k_{ij} regression. Freshwater = converged points used for k_{ij}^{AQ} regression; Brine = matched fresh/brine pairs retained for Sechenov validation after source exclusions.

Gas	Total Points	Freshwater	Brine	T Range (K)	P Range (bar)
CO_2	716	668	48	273–553	0.4–2000
H_2S	523	480	43	294–598	1–309
CH_4	175	143	32	298–507	22–1925
H_2	190	162	28	273–483	3.5–1013
N_2	173	134	39	298–513	3.5–1013
C_2H_6	111	111	—	275–573	0.5–3500
C_3H_8	59	59	—	279–422	5–192
nC_4H_{10}	27	27	—	411–511	14–689
Total	**1,974**	**1,784**	**190**		

5. Results: Freshwater BIP Correlations

5.1. Overview of Functional Forms

Different gases exhibit distinct k_{ij}^{AQ} versus temperature behaviours, requiring gas-specific functional forms. Based on the pointwise regression results, we adopt:

- **Cubic polynomial in T_r** (CO_2): reproduces the non-linear increase with accelerating rise at high T.
- **Exponential form in T_r** (H_2S): reproduces the strong temperature dependence and asymptotic behaviour.
- **Rational form in T_r** (CH_4, H_2, C_2H_6, C_3H_8): reproduces non-linear temperature dependence with bounded high-T extrapolation.
- **Linear in T_r** (N_2): adequate for the near-linear behaviour observed.
- **Quadratic in T_r** (nC_4H_{10}): original S&W generalised HC form retained.

These are the forms of the published version, re-used unchanged. Section 5.10 and Appendix Appendix C test whether they remain adequate on the coupled procedure. Several coefficient sets below differ substantially from the published values while the correlated curves move by at most 0.030 over the fitted range for five of the seven gases (H_2S 0.067, H_2 0.066, in each case at the highest temperatures), because the parameters of the exponential and rational forms are strongly correlated. Use all digits given.

5.2. CO_2 BIP Correlation

Figure 1a shows the pointwise-regressed k_{ij}^{AQ} values for CO_2 compared against the proposed correlation and S&W original. The CO_2 dataset is the largest (668 freshwater points from 24 sources spanning 1855–2021). Figure 3a shows the corresponding non-aqueous phase BIP.

The proposed correlation is:

$$k_{ij,\mathrm{CO_2}}^{\mathrm{AQ}} = -1.550211 + 2.835971\,T_r - 1.830119\,T_r^2 + 0.430056\,T_r^3 \tag{17}$$

where $T_r = T/T_c$ with $T_c = 304.20$ K. This achieves MAE = 0.0087 versus the pointwise values (606 freshwater points), compared to MAE = 0.0120 for the original S&W correlation (Eq. 8 at $c_{\mathrm{sw}} = 0$).

5.3. H_2S BIP Correlation

Figure 1b shows the H_2S aqueous-phase BIP results; Figure 3b shows the non-aqueous phase BIP.

The proposed correlation is:

$$k_{ij,\mathrm{H_2S}}^{\mathrm{AQ}} = -\frac{0.22181}{T_r} + 1.73949 \times 10^6 \exp\left(-\frac{23.00392}{T_r}\right) + 0.25183 \tag{18}$$

where $T_r = T/T_c$ with $T_c = 373.20$ K. This achieves MAE = 0.0070 versus the pointwise values (405 freshwater points), compared to MAE = 0.0079 for the original S&W correlation (Eq. 9). The exponential term is negligible below about 100°C and supplies the upturn at 150–200°C.

5.4. CH_4 BIP Correlation

Figure 1e shows the CH_4 results. The proposed correlation uses the rational form:

$$k_{ij,\mathrm{CH_4}}^{\mathrm{AQ}} = \frac{-2.197678 + T_r}{0.7150154 + 0.894503\,T_r} \tag{19}$$

where $T_r = T/T_c$ with $T_c = 190.60$ K. This achieves MAE = 0.0076 over 115 freshwater points (T ≤ 200°C), the same as the original S&W correlation (0.0076).

5.5. N_2 BIP Correlation

Figure 1c shows the N_2 results. A simple linear correlation suffices:

$$k_{ij,\mathrm{N_2}}^{\mathrm{AQ}} = -1.621096 + 0.415534\,T_r \tag{20}$$

where $T_r = T/T_c$ with $T_c = 126.10$ K. This achieves MAE = 0.0104 over 127 points (S&W: 0.0132).

5.6. H_2 BIP Correlation

Hydrogen represents a new extension to the S&W framework. Data quality assessment reveals systematic inter-source disagreements of 10–17% at identical conditions, and qualitative disagreement on the temperature dependence: Wiebe & Gaddy [15], Chabab et al. [13], and Torín-Ollarves & Trusler [14] show U-shaped solubility with minimum near 50°C, whilst other sources show monotonic increase. Three sources demonstrating consistent U-shaped behaviour were selected for correlation development: Wiebe (IUPAC-evaluated, 0–100°C), Chabab 2023 (modern high-pressure, 25–50°C), and T-O 2021 (50–150°C).

A linear BIP correlation cannot reproduce the U-shaped solubility; a rational form is required:

$$k_{ij,\mathrm{H_2}}^{\mathrm{AQ}} = \frac{-16.01983 + T_r}{0.7312363 + 0.681362\,T_r} \tag{21}$$

where $T_r = T/T_c$ with $T_c = 33.145$ K. The correlation is monotonic in T_r; the solubility minimum emerges from the EOS calculation at the fitted BIP values, with the rational form supplying the curvature in $k_{ij}(T)$ that a linear form cannot. Across 154 freshwater points (T ≤ 200°C) the correlation achieves MAE(k_{ij}) = 0.035, with a solubility MARE of 11.3% over the 142 points carrying measured solubility data. H_2 is the one gas for which the refit scores worse than the published dry-vapour fit (10.2%); the difference is at 100–180°C, where the data are sparse (47 points) and the pointwise scatter is the largest of any gas.

5.7. C_2H_6 and C_3H_8 BIP Correlations

For ethane and propane, the rational form reproduces the observed temperature dependence:

Ethane:

$$k_{ij,\mathrm{C_2H_6}}^{\mathrm{AQ}} = \frac{-1.271348 + T_r}{-0.02927137 + 1.637183\,T_r} \tag{22}$$

where $T_r = T/T_c$ with $T_c = 305.40$ K. Fitted to 95 freshwater points, achieving MAE = 0.0063 (S&W: 0.0086).

Propane:

$$k_{ij,\mathrm{C_3H_8}}^{\mathrm{AQ}} = \frac{-1.156778 + T_r}{0.208946 + 1.810663\,T_r} \tag{23}$$

where $T_r = T/T_c$ with $T_c = 369.80$ K. Fitted to 59 freshwater points, achieving MAE = 0.0015 (S&W: 0.0041).

Figures 1f and 1g show the C_2H_6 and C_3H_8 results respectively.

5.8. n*-Butane BIP Correlation*

For $n\mathrm{C_4H_{10}}$, only 27 freshwater data points are available, which is insufficient for independent correlation development. The original S&W generalised hydrocarbon form (Eq. 6 at $c_{\mathrm{sw}} = 0$) is retained:

$$k_{ij,\mathrm{nC_4H_{10}}}^{\mathrm{AQ}} = -0.9354 + 1.2615\,T_r - 0.3696\,T_r^2 \tag{24}$$

where $T_r = T/T_c$ with $T_c = 425.2$ K ($\omega = 0.193$). S&W fitted this form on the same equilibrium procedure, so it carries over unchanged.

5.9. Summary of Aqueous-Phase BIP Correlations

Figure 1 provides an overview of the gases with aqueous data, showing pointwise-regressed k_{ij}^{AQ} values versus temperature alongside the original S&W correlations, the published (dry-vapour) correlations and the refit. Table 6 summarises the proposed correlations with performance metrics. Full per-source data attribution and reproducible regression scripts are available in the project repository [22].

For heavier hydrocarbons ($i\mathrm{C_4H_{10}}$ and above), the original S&W parameterisation based on acentric factor and critical temperature (Eq. 6) is retained; no new data were compiled for these species. S&W fitted it on the same procedure, so it is consistent with the refit.

5.10. Adequacy of the Functional Forms on the S&W Procedure

The forms were selected on the published dry-vapour pointwise values. Two checks test whether they still describe the pointwise values on the coupled procedure. First, each adopted form was compared with a five-parameter quartic in T fitted by the same L1 criterion to the same points (Table 7). The quartic can follow any smooth temperature trend the adopted form misses, so the gap between the two MAEs bounds the error the form itself contributes. The gap is at most 0.0008 on the coupled procedure (H_2, 2% of its MAE) and 0.0005 on the dry-vapour basis; relative to each gas's MAE it is at most 7% on the coupled procedure (C_3H_8, 0.0001 of 0.0015) against at most 22% on the dry-vapour basis (C_3H_8). The gap grows only for N_2 (0.0002 to 0.0004) and H_2 (0.0004 to 0.0008). Second, the binned residuals (Figure 2) show no temperature trend that the published residuals did not already have, with one exception: H_2 at 140–160°C, where the median residual is +0.048 (published +0.019) on 8 points, followed by −0.015 at 160–180°C

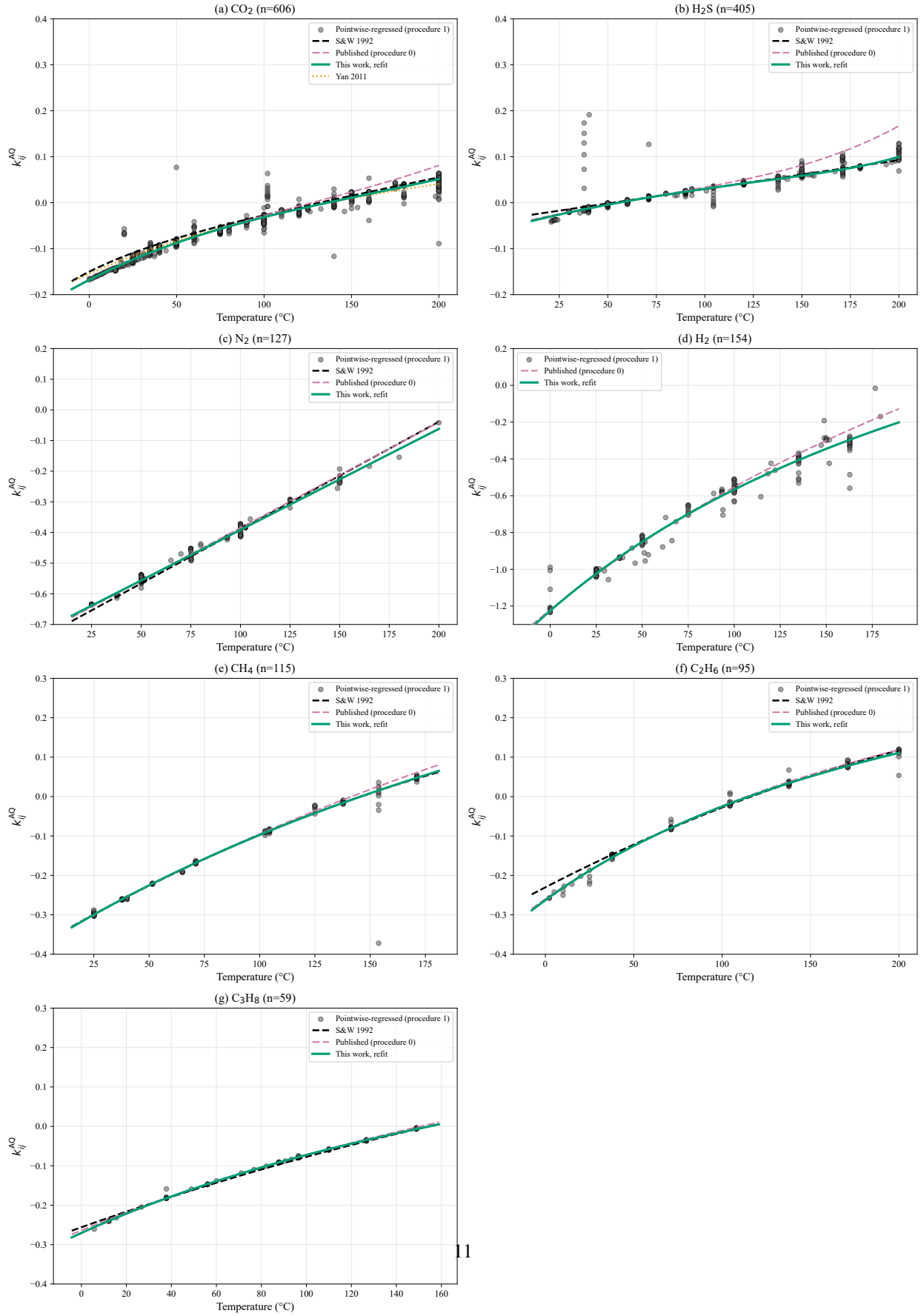


Figure 1: Overview of aqueous-phase BIP correlations for the eight gases with aqueous data (nC_4H_{10} omitted; no data). Grey symbols: pointwise-regressed k_{ij}^{AQ} values on the coupled procedure (Eq. 10), pooled across sources. Black dashed lines: original S&W correlations; pink dashed: published dry-vapour correlations [34]; green solid lines: refit correlations (this work); orange dotted: Yan et al. [19] (CO_2 only).

Table 6: Summary of proposed aqueous-phase BIP correlations. Error metrics defined in Eq. 15: MAE(k_{ij}) is evaluated against pointwise-regressed values; MARE(%) is evaluated on predicted solubility x_{gas}. Both MARE columns use the coupled procedure (Eq. 10); the S&W column uses the original correlation at zero salinity with the S&W k_{ij}^{NA} (Eq. 17 for H_2S). CO_2: five Wang 2021 points excluded (Section 4).

Gas	Form	n	MAE(k_{ij})	MARE(%)	S&W MARE(%)	Eq.
CO_2	Cubic in T_r	606	0.0087	12.4	15.3	17
H_2S	Exponential in T_r	405	0.0070	31.9[a]	30.8	18
CH_4	Rational	115	0.0076	3.9	3.9	19
N_2	Linear in T_r	127	0.0104	5.3	7.2	20
H_2	Rational	154[b]	0.0350	11.3	—	21
C_2H_6	Rational	95	0.0063	8.4	11.3	22
C_3H_8	Rational	59	0.0015	3.4	8.9	23
nC_4H_{10}	S&W quadratic[c]	27	—	—	—	24

[a] 6.3% (S&W 7.8%) excluding the 20 low-x points ($x < 0.003$); see Section 8.7.

[b] MAE(k_{ij}) over 154 freshwater points (T ≤ 200°C); the 10.2% solubility MARE is over 142 points (12 water-content-only Gillespie (1980) [28] points excluded).

[c] Original S&W generalised HC form (Eq. 6) retained; limited data.

on 18 points. A rational form cannot follow a step of that size over 20°C, and the two bins are dominated by different sources (Suciu 1951 and Torín-Ollarves 2021 at 140–160°C, Stephan 1956 at 160–180°C); the pattern is consistent with inter-source disagreement, which no smooth $k_{ij}(T)$ form would remove. H_2S has wide interquartile bands at 100–120°C and 160–180°C, with bin medians within ±0.004.

Table 7: Form adequacy: L1 MAE(k_{ij}^{AQ}) of the adopted form and of a five-parameter quartic in T fitted to the same pointwise values, on both bases. Gap = adopted − quartic. Dry-vapour basis: published pointwise values and correlations [34], on the points that converge on both bases.

			Dry vapour (published)			Coupled (this work)		
Gas	Form	n	Adopted	Quartic	Gap	Adopted	Quartic	Gap
CO_2	cubic	606	0.0114	0.0113	0.0001	0.0087	0.0086	0.0001
H_2S	exponential	405	0.0112	0.0110	0.0002	0.0070	0.0070	0.0000
N_2	linear	127	0.0114	0.0112	0.0002	0.0104	0.0100	0.0004
H_2	rational	154	0.0300	0.0296	0.0004	0.0350	0.0342	0.0008
CH_4	rational	115	0.0089	0.0088	0.0001	0.0076	0.0076	0.0001
C_2H_6	rational	95	0.0095	0.0090	0.0005	0.0063	0.0061	0.0003
C_3H_8	rational	59	0.0023	0.0018	0.0005	0.0015	0.0014	0.0001

5.11. Non-Aqueous Phase BIP (k_{ij}^{NA})

The non-aqueous phase BIP governs the predicted water content of the gas phase, y_{H_2O}. For each gas, pointwise k_{ij}^{NA} values were regressed from experimental y_{H_2O} data with Eq. 10 and the refit k_{ij}^{AQ} in the aqueous phase, and the optimal constant k_{ij}^{NA} was determined by minimising the mean absolute relative error (MARE) on y_{H_2O} predictions via two-stage grid search (coarse 0.02, fine 0.001). Data were filtered to exclude pressures where y_{H_2O} is insensitive to k_{ij}^{NA} ($P < 20$ bar; $P < 50$ bar for H_2), and points where the regression hit search bounds.

Figure 3 shows the pointwise-regressed k_{ij}^{NA} values for the seven gases with data alongside the original S&W values and proposed recommendations.

Table 8 presents the complete set of recommended k_{ij}^{NA} values. The original S&W Table 5 constants are retained for every gas that has them: on the coupled procedure, re-optimisation improves the y_{H_2O} MARE by at most 0.9 percentage points (C_2H_6; Table 8), below the 1.1 pp that the published version treated as negligible. For H_2S the published version replaced the temperature-dependent S&W form (Eq. 17: $k_{ij}^{\text{NA}} = 0.19031 - 0.05965\,T_r$) with the constant 0.161, a 3.1 pp gain on the dry-vapour basis. On the coupled procedure the gain is 0.4 pp (7.1% to 6.7%, $n = 162$), which by the same rule would not justify the change; 0.161 is retained for continuity with the published version. The pointwise k_{ij}^{NA} values for H_2S scatter widely (standard deviation 0.34) and do not support a temperature-dependent fit.

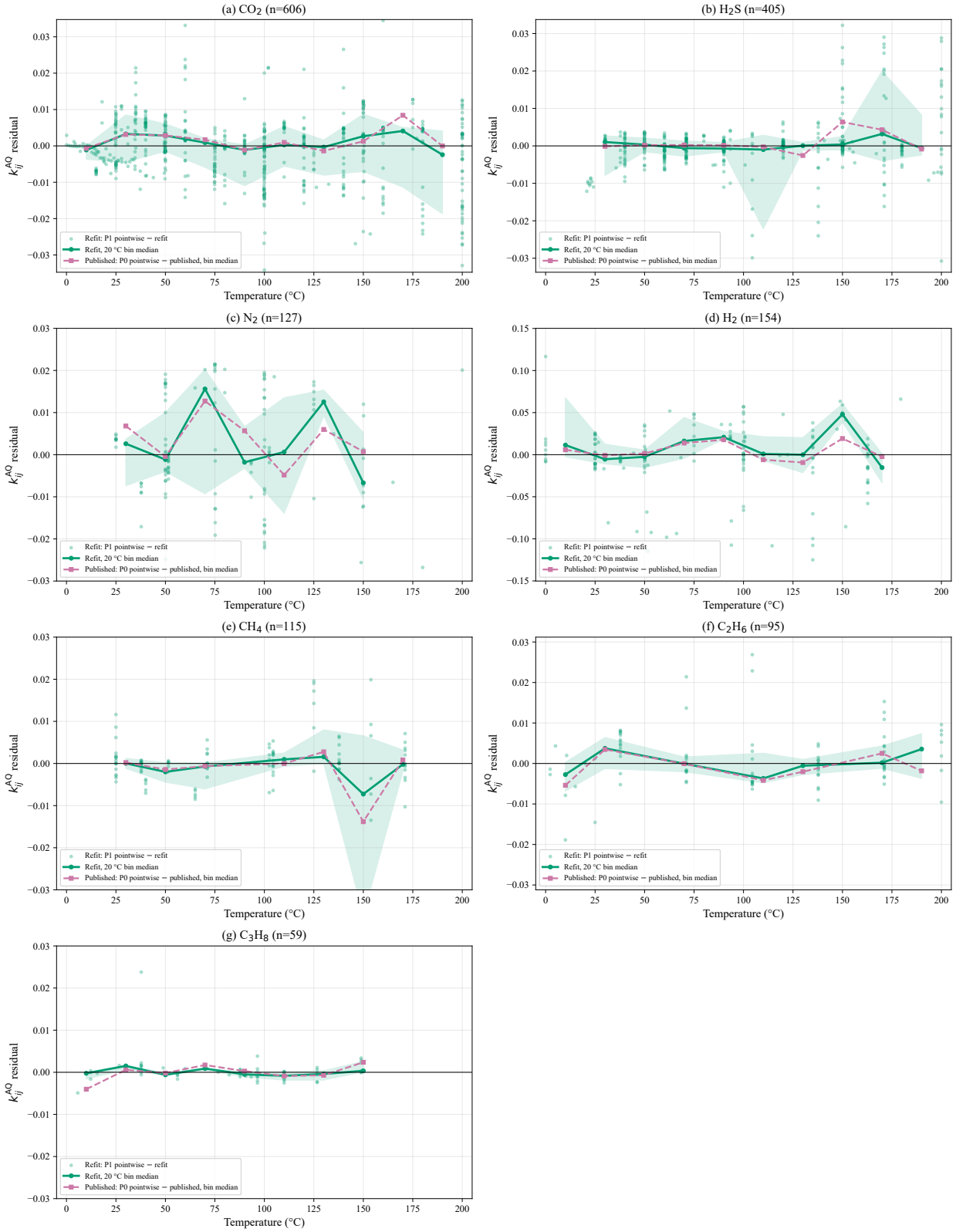


Figure 2: Residuals of the pointwise k_{ij}^{AQ} about the correlations. Green symbols: pointwise value on the coupled procedure minus the refit correlation; green line and band: 20°C bin median and interquartile range (bins with at least 3 points). Pink dashed line: bin median of the published dry-vapour residuals [34], for comparison. The vertical scale clips the outliers that dominate the H_2S MARE (Section 8.7).

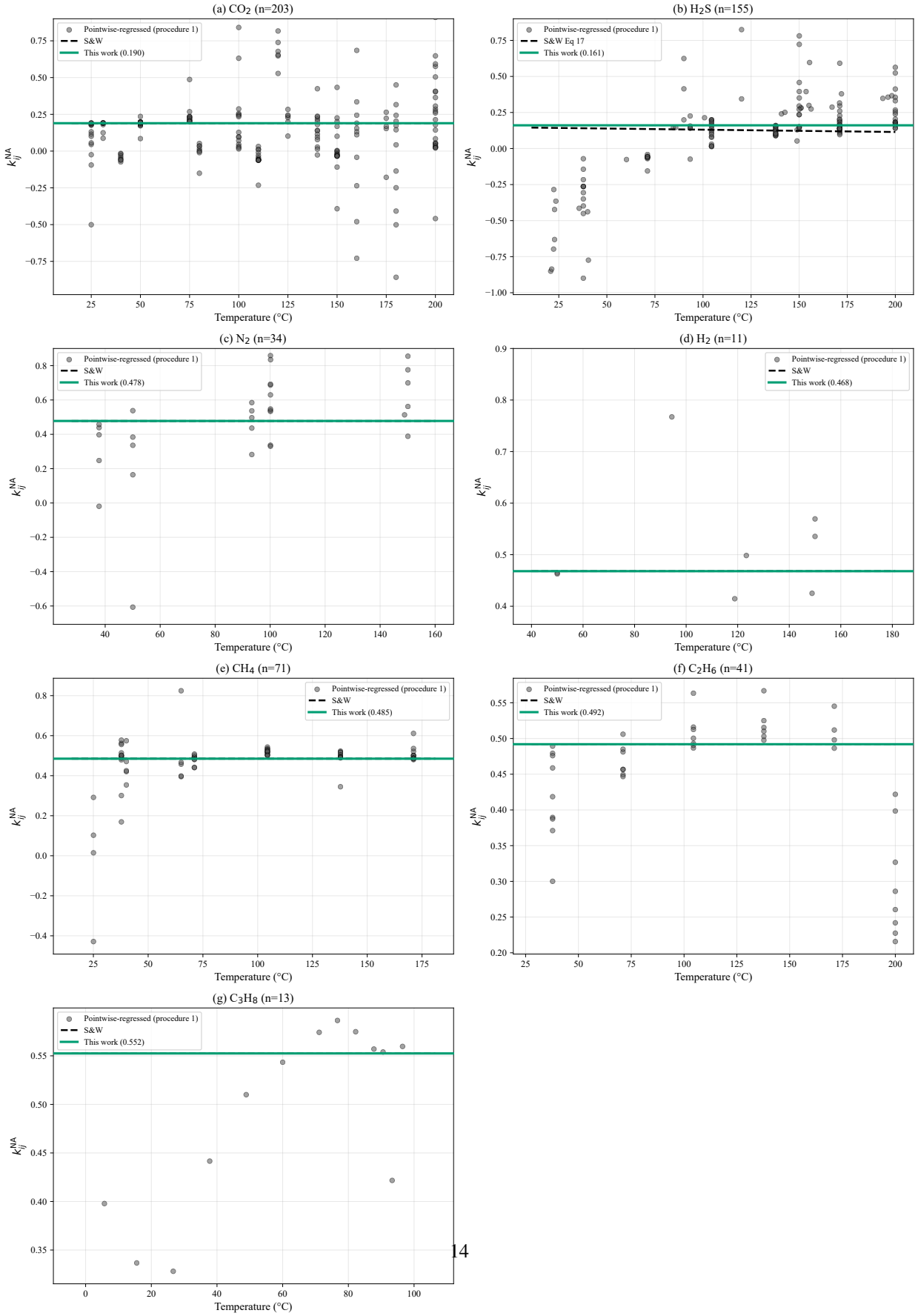


Figure 3: Overview of non-aqueous phase BIP for the seven gases with non-aqueous data (nC$_4$H$_{10}$ omitted; no data). Grey symbols: pointwise-regressed k_{ij}^{NA} from experimental $y_{\mathrm{H_2O}}$ data on the coupled procedure with the refit k_{ij}^{AQ} (pooled across sources; the points of the published figure that converge on this basis). Black dashed lines: original S&W values (Eq. 17 for H$_2$S; Table 5 constants for others). Green solid lines: recommended values (this work).

Table 8: Recommended non-aqueous phase BIP values. n: quality-filtered data points (bound-hitting and low-P excluded). MARE (Eq. 15) evaluated on y_{H_2O} predictions with Eq. 10 and the refit k_{ij}^{AQ}. Boldface indicates changed values relative to S&W 1992.

Gas	n	S&W k_{ij}^{NA}	S&W MARE(%)	Recommended	MARE(%)	Optimum (MARE)[d]
CO_2	208	0.1896	31.6	0.1896	31.6	0.190 (31.6)
H_2S	**162**	**Eq. 17**[a]	**7.1**	**0.161**	**6.7**	0.147 (6.4)
CH_4	93	0.4850	7.1	0.4850	7.1	0.495 (6.7)
N_2	31	0.4778	8.1	0.4778	8.1	0.537 (7.5)
H_2	14	—[b]	—	0.468	4.2	0.499 (4.1)
C_2H_6	58	0.4920	19.0	0.4920	19.0	0.476 (18.1)
C_3H_8	11	0.5525	5.5	0.5525	5.5	0.557 (5.4)
nC_4H_{10}	0	0.5091	—	0.5091	—	—[c]

[a] S&W Eq. 17: $k_{ij}^{NA} = 0.19031 - 0.05965\,T_r$ where $T_r = T/T_{c,H_2S}$. Replaced by constant 0.161 (published version).

[b] New in this work (H_2 not in original S&W); validated at $P \geq 50$ bar.

[c] No quality-filtered y_{H_2O} data; the S&W constant is adopted unchanged.

[d] Best constant from the two-stage grid search and its MARE, shown to support retention.

For H_2, the constant $k_{ij}^{NA} = 0.468$ is consistent with the volatility trend established by CH_4 (0.485) and N_2 (0.478) when extrapolated to H_2's boiling point ($T_b = -253$°C). Quality filtering to $P \geq 50$ bar yields a MARE of 4.2% on 14 points.

The high CO_2 MARE (32%) and C_2H_6 MARE (19%) reflect a structural limitation: the data for these gases span temperatures both below and above the gas critical point (CO_2: $T_c = 304$ K; C_2H_6: $T_c = 305$ K), across which vapour-phase behaviour differs and a single constant BIP cannot represent both regimes. Nevertheless, re-optimisation provides negligible improvement, indicating the limitation is structural rather than parametric.

Scatter in pointwise k_{ij}^{NA}.. The large scatter visible in Figure 3 is largely an artefact of regressing a second-order parameter from noisy measurements. At low to moderate pressures, y_{H_2O} is dominated by the vapour pressure of water (the Raoult's law contribution), which accounts for 90–97% of the total water content below 50 bar. The k_{ij}^{NA} parameter affects only the fugacity coefficient correction to this dominant term. A 5% measurement error in y_{H_2O} at 20 bar can propagate to a k_{ij}^{NA} uncertainty exceeding ±0.5, whereas the same measurement error at 200 bar constrains k_{ij}^{NA} to ±0.05. This pressure-dependent sensitivity amplification explains why low-P data produce extreme scatter while high-P data cluster tightly around the recommended constants.

Volatility trend.. Despite this scatter, the recommended k_{ij}^{NA} values exhibit a coherent pattern when plotted against normal boiling point T_b. The six non-acid gases (H_2, N_2, CH_4, C_2H_6, C_3H_8, nC_4H_{10}) rise from 0.468 to 0.51–0.55 with increasing T_b, while the two acid gases (CO_2: 0.190; H_2S: 0.161) plot distinctly lower. This grouping provides a consistency check: constants fitted individually from noisy y_{H_2O} data recover a coherent pattern rather than random scatter.

6. Results: Salting-Out Models

6.1. Evaluation Methodology

We extracted 573 experimental Sechenov coefficients from matched freshwater/brine pairs in the compiled dataset, distributed as: CO_2 (69), H_2S (239), CH_4 (198), N_2 (39), and H_2 (28). Of these, the points from the sources excluded in Section 4 (Prutton & Savage [26] for CO_2; Blount & Price [24] and McGee [25] for CH_4; Barrett [27] for H_2S) are retained in the figures but omitted from the quantitative validation. For each matched pair at identical temperature and pressure:

$$k_{s,\mathrm{exp}} = \frac{\log_{10}(x_{\mathrm{fresh}}/x_{\mathrm{brine}})}{m} \tag{25}$$

These experimental values were compared against: (1) the S&W Sechenov correlation (Eq. 13); (2) gas-specific Pitzer models from the literature; and (3) implied k_s from embedded BIP correlations. Figure 4 shows the five gases for which experimental brine data or gas-specific Pitzer models exist.

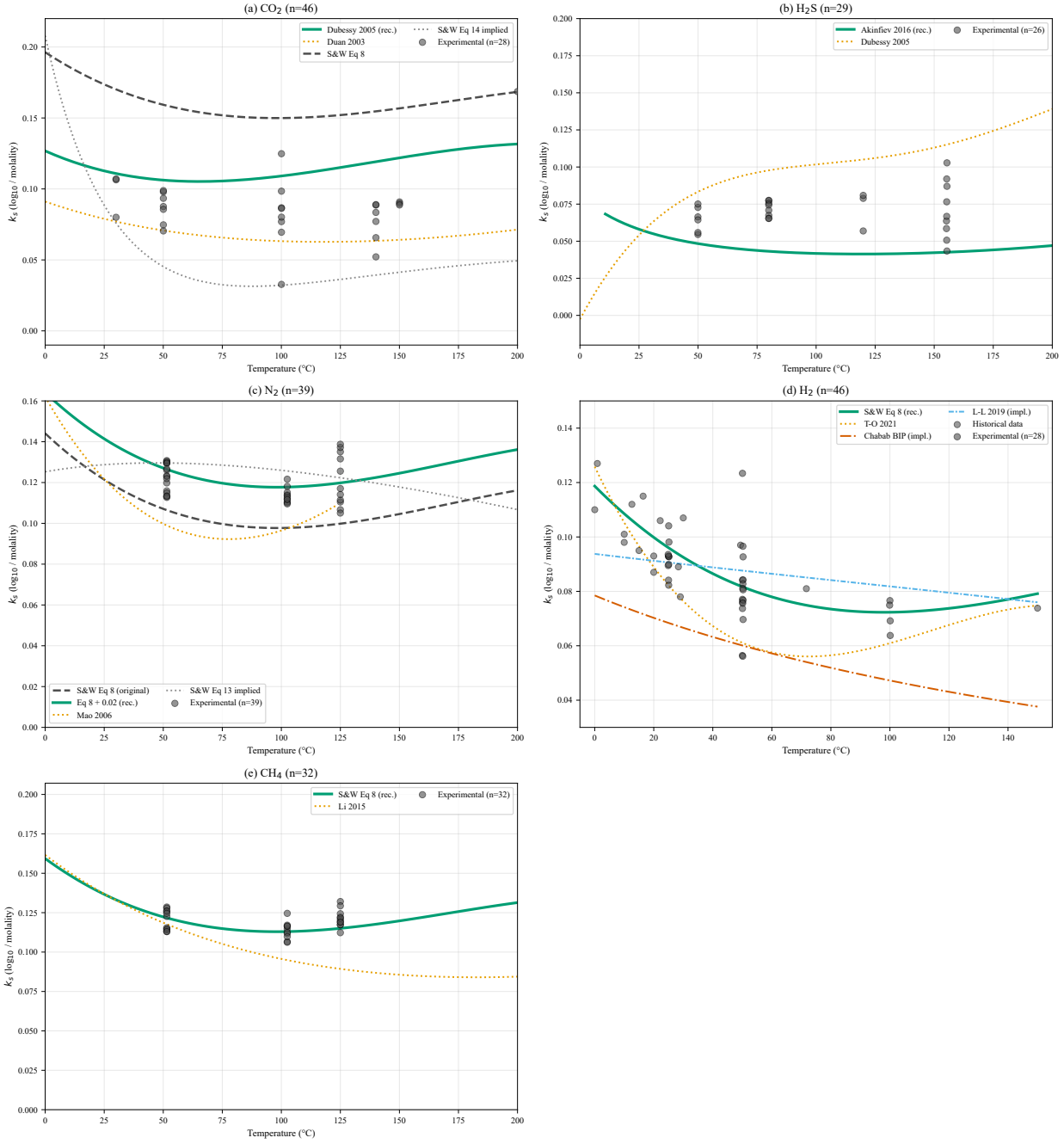


Figure 4: Sechenov coefficients (k_s) for the five gases with available brine data or gas-specific Pitzer models. Grey symbols: experimental k_s extracted from matched freshwater/brine pairs (Eq. 25), pooled across sources. Green solid lines: the recommended model. Coloured dotted/dash-dot lines: alternative published models. Black dashed lines: the S&W Sechenov correlation (Eq. 13), shown for reference; for CH_4 and H_2 the recommendation *is* S&W Eq. 13, drawn in the green recommended style. For C_2H_6, C_3H_8, and nC_4H_{10}, no experimental brine data at reservoir conditions exist and the S&W correlation is retained (Section 6.7). Implied k_s of the Chabab and Lopez-Lazaro BIPs (panel d) are evaluated with Eq. 10 at 100 bar, m = 1.5 mol/kg; models and data are otherwise independent of the equilibrium procedure.

6.2. CO_2 Salting-Out

Figure 4a compares experimental CO_2 Sechenov coefficients against model predictions.

The original S&W framework embedded CO_2 salinity effects directly in the BIP correlation (Eq. 8), a practical approach given the data available in 1992. When the S&W Sechenov correlation (Eq. 13) is instead applied to CO_2 using its boiling point (T_b = 194.7 K), it overpredicts salting-out by approximately 2× (MAE = 0.076), confirming that this hydrocarbon-derived correlation is not suitable for acid gases. The Dubessy et al. (2005) [5] extended Sechenov model agrees best with the experimental k_s values (MAE = 0.030); a constant offset of −0.011 improves the k_s MAE to 0.020 and the brine solubility MARE from 18.7% to 17.3% (n = 109, explicit Sechenov on the refit freshwater baseline). The offset was chosen in the published version to minimise brine MARE on the dry-vapour baseline; on the coupled procedure the optimum is −0.016 (17.1%), a 0.2 pp difference, so −0.011 is retained. The Duan & Sun (2003) [4] Pitzer model, with explicit pressure and molality dependence, gives comparable agreement with the experimental k_s values (MAE = 0.023) but a higher brine solubility MARE (23.5% versus 17.3% for the offset Dubessy model, n = 109).

Recommendation: Use Dubessy et al. [5] with a constant offset of −0.011 kg/mol for CO_2-brine systems.

6.3. H_2S Salting-Out

Figure 4b compares experimental H_2S Sechenov coefficients against model predictions.

The original S&W framework provided no salting-out model for H_2S: the BIP correlation (Eq. 9) contains no salinity terms, so H_2S-brine calculations could not be performed within the standard S&W approach. If the S&W Sechenov correlation (Eq. 13) were applied using the H_2S boiling point (T_b = 212.8 K), it would overpredict salting-out by approximately 3× (MAE = 0.096 over the 29 retained matched pairs), with predicted $k_s \approx 0.158$ at 100°C versus experimental $k_s \approx 0.05$ kg/mol. The Akinfiev et al. (2016) [6] Pitzer model, parameterised to 700+ experimental points, agrees with the same 29 matched pairs to MAE = 0.024. Applying a constant upward shift of +0.019 kg/mol, calibrated in the published version to minimise brine solubility MARE, improves the k_s MAE to 0.013 and reduces the brine MARE from 19.0% to 10.0% (explicit Sechenov on the refit freshwater baseline, n = 100; the embedded implementation gives 11.0%). On the coupled procedure the MARE-optimal shift is +0.017 (9.9%), so +0.019 is retained.

Recommendation: Use Akinfiev et al. [6] with a constant offset of +0.019 kg/mol for H_2S-brine systems in the range 283–573 K.

6.4. N_2 Salting-Out

Figure 4c shows N_2 Sechenov coefficients. The 39 experimental k_s values available (O'Sullivan & Smith [23]) span 52–125°C at m = 1 and 4 mol/kg. The unmodified S&W Sechenov correlation (Eq. 13) slightly underpredicts the experimental data and diverges from the Mao & Duan (2006) [12] Pitzer model at higher temperatures. Applying a constant upward shift of +0.02 to Eq. 13 improves the k_s MAE from 0.018 to 0.007 and tracks the Mao & Duan model across the temperature range. **Recommendation:** Use a modified Eq. 13 for N_2:

$$k_{s,\mathrm{N_2}}(T) = k_{s,\text{Eq. 13}}(T) + 0.02 \tag{26}$$

with Mao & Duan (2006) as an alternative for temperatures below 400 K.

6.5. CH_4 Salting-Out

Figure 4e shows CH_4 Sechenov coefficients. Three brine CH_4 solubility datasets were evaluated: O'Sullivan & Smith (1970) [23] (32 points, m = 1–4 mol/kg, 51–125°C), Blount & Price (1982) [24] (773 points, m = 0.06–7.1 mol/kg, 23–240°C), and McGee et al. (1981) [25] (23 points, 212–292°C).

Blount & Price data were excluded from validation based on three diagnostic indicators of systematic bias: (i) implied k_s values collapse from ~0.075 at $m \leq 1$ to ~0.044 at m = 4–7, violating the Sechenov linearity expected below 6 mol/kg; (ii) at $m < 0.2$, the data imply negative k_s (salting-*in*), which is inconsistent with all other CH_4 brine data and with the Sechenov framework for NaCl solutions; (iii) at overlapping conditions (T = 50–125°C, m = 1–4), Blount implied $k_s \approx 0.054$ compared to $k_s \approx 0.119$ from O'Sullivan, a 2.2× discrepancy. McGee et al. were excluded because their freshwater baseline already deviates by MARE = 52% from our model, making brine evaluation unreliable.

The retained O'Sullivan data (n = 32) yield implied k_s = 0.106–0.132 kg/mol across m = 1–4 with standard deviation 0.007, consistent with the S&W Sechenov correlation (Eq. 13) prediction of $k_s \approx 0.12$–0.13 in the same temperature range. The corresponding brine parity MARE is 6.5% with the embedded correction and 6.7% with the explicit Sechenov correction (Figure A.7). The Li et al. (2015) [7] Pitzer model agrees with S&W Eq 13 at 25°C ($k_s \approx$ 0.137) but predicts lower values at higher temperatures (k_s = 0.086 at 200°C vs S&W 0.131). **Recommendation:** Use S&W Eq 13 for CH_4, validated by O'Sullivan data; Li et al. as an alternative below 100°C.

6.6. H_2 Salting-Out

Figure 4d compares H_2 Sechenov coefficients from six experimental sources spanning 1952–2023 against the S&W Sechenov correlation (Eq. 13) and several published BIP formulations.

No Pitzer parameterisations were found for H_2-NaCl-H_2O. The S&W Sechenov correlation (Eq. 13) with T_b = 20.3 K produces the lowest k_s of any gas in the framework and agrees with 46 experimental points from six independent sources with MAE = 0.0087 kg/mol. Despite being developed in 1992 for C_1–C_4 hydrocarbons with boiling points 112–273 K, the correlation reproduces hydrogen data (T_b = 20.3 K) to within 2% of the Chabab et al. (2023) measurement at the correlation minimum near 100°C; agreement at lower temperatures is less precise due to scatter among historical sources.

The Torín-Ollarves & Trusler (2021) polynomial correlation, whilst fitted to modern high-pressure data, is anchored primarily to two directly measured temperatures (50 and 150°C, where their pure-water reference measurements exist), with all brine data at a single salinity of 2.5 mol/kg. The T-O correlation at 50°C ($k_s \approx 0.061$ kg/mol) appears anomalously low compared to three independent sources at similar temperatures (Morrison & Billett [16]: 0.097; Chabab 2023 [13]: 0.077; Gerecke & Bittrich [17]: 0.076 kg/mol).

Figure 4d also shows implied k_s values from BIP formulations that embed salinity effects directly. The Chabab et al. [13] BIP yields implied k_s values 15–30% lower than experimental measurements with monotonic decline, missing the characteristic minimum near 100°C. Lopez-Lazaro et al. [18] shows different temperature behaviour, with implied k_s overpredicting at high temperatures. In contrast, the embedded BIP from this work closely tracks the S&W Sechenov correlation (Eq. 13) by construction.

Recommendation: Use the S&W Sechenov correlation (Eq. 13) with T_b = 20.3 K for H_2. Additional experimental validation above 100°C would strengthen confidence in the predicted upturn above the minimum.

6.7. Heavier Hydrocarbons

We found no experimental measurements of C_2H_6, C_3H_8, or nC_4H_{10} solubility in NaCl brines at reservoir temperatures and pressures published since the original S&W framework. The Li et al. [7] Pitzer parameterisations for these gases are fitted to 1-atm brine data at near-ambient temperatures and exhibit unreliable extrapolation at higher temperatures, predicting near-zero or negative k_s values above 200°C. Recent molecular simulation results (Fang et al. 2023 [8]) provide computed solubilities of C_1–nC_4 in NaCl solutions at 278–308 K and 1–300 bar, but remain limited to near-ambient temperatures. In the absence of validated data at reservoir conditions, we retain the S&W Sechenov correlation (Eq. 13) for C_2H_6 (T_b = 184.6 K), C_3H_8 (T_b = 231.1 K), and nC_4H_{10} (T_b = 272.7 K). **Recommendation:** Use Eq. 13 for heavier hydrocarbons pending new experimental data at elevated temperatures.

6.8. Summary of Salting-Out Recommendations

Table 9 summarises the recommended Sechenov models.

7. Results: Embedded Salinity BIP

7.1. Motivation

For commercial simulators where flash algorithm modification is impractical, the salting-out effect can be embedded directly into the aqueous BIP:

$$k_{ij}^{\mathrm{AQ}}(T, m) = k_{ij,\mathrm{fw}}(T) + \Delta k_{ij}(T, m) \tag{27}$$

where $k_{ij,\mathrm{fw}}(T)$ is the freshwater BIP and $\Delta k_{ij}(T, m)$ is the salinity correction term.

Table 9: Recommended Sechenov coefficient models by gas.

Gas	Recommended Model	MAE vs Data	Original S&W Treatment
CO_2	Dubessy et al. [5]−0.011	0.020	Embedded in BIP (Eq. 8)
H_2S	Akinfiev et al. [6]+0.019	0.013	No salting-out (Eq. 9 freshwater only)
CH_4	Eq. 13	0.006*	Eq. 13 via Eq. 6 (diverges at high T)
N_2	Eq. 13 + 0.02	0.007	Embedded in BIP (Eq. 7)
H_2	Eq. 13 (T_b = 20.3 K)	0.009	Not in original framework
C_2H_6, C_3H_8, nC_4H_{10}	Eq. 13†	—	Eq. 13 via Eq. 6

*O'Sullivan & Smith (1970) data only; Blount & Price (1982) excluded due to systematic bias (see Section 6.5).
†No experimental brine data at reservoir conditions; S&W correlation retained (Section 6.7).

7.2. Functional Forms

For every gas except CO_2, the salinity correction takes the linear-in-m form (3 parameters):

$$\Delta k_{ij} = (a_0 + a_1 T_r + a_2 T_r^2) \cdot m \tag{28}$$

where $T_r = T/T_c$ is the reduced temperature of the gas. For CO_2, a quadratic-in-m extension (5 parameters) provides improved accuracy at higher salinities:

$$\Delta k_{ij} = (a_0 + a_1 T_r + a_2 T_r^2) \cdot m + (b_0 + b_1 T_r) \cdot m^2 \tag{29}$$

The quadratic term improves fit fidelity to the Dubessy et al. (2005) model at $m > 2$ mol/kg; in the published fit it reduced the CO_2 solubility MAE from 3.1% (linear-in-m) to 1.56%, and it is retained here (2.92% on the coupled procedure).

7.3. Fitted Coefficients

Coefficients were fitted by minimising the solubility error against the explicit Sechenov target on a grid of temperature, pressure and molality, with the refit freshwater k_{ij}^{AQ} and Eq. 10 (brine S&W water alpha in both phases). The target is the published definition: the freshwater solubility at the same T and total P, multiplied by $10^{-k_s m}$. Defining it instead as an activity coefficient at equal gas fugacity changes every brine MARE in Table C.17 by at most 0.3 pp. Table 10 presents the fitted values. CO_2 uses the quadratic form (Eq. 29); all other gases use the linear form (Eq. 28). C_3H_8 and nC_4H_{10} are now fitted on their own freshwater BIPs; the published version carried them over from the MC-3 fit.

Table 10: Embedded salinity BIP coefficients (coupled procedure). CO_2 uses Eq. 29 (5 parameters); all others use Eq. 28 (3 parameters). MAE is on predicted solubility relative to the explicit Sechenov target over the fitting grid.

Gas	T_c (K)	a_0	a_1	a_2	b_0, b_1	MAE (%)
CO_2†	304.20	0.040848	−0.078986	0.050276	0.006766, −0.007356	2.92
H_2S	373.20	0.047438	−0.091614	0.049824	—	2.35
CH_4	190.60	0.134391	−0.134258	0.040843	—	0.98
N_2	126.10	0.219484	−0.148600	0.030617	—	1.04
H_2	33.145	0.376712	−0.0647304	0.00315541	—	0.66
C_2H_6	305.40	0.077770	−0.123360	0.062587	—	1.09
C_3H_8	369.80	0.058057	−0.111055	0.069227	—	1.40
nC_4H_{10}	425.20	0.045642	−0.100089	0.072927	—	1.75

†Quadratic-in-m form (Eq. 29).

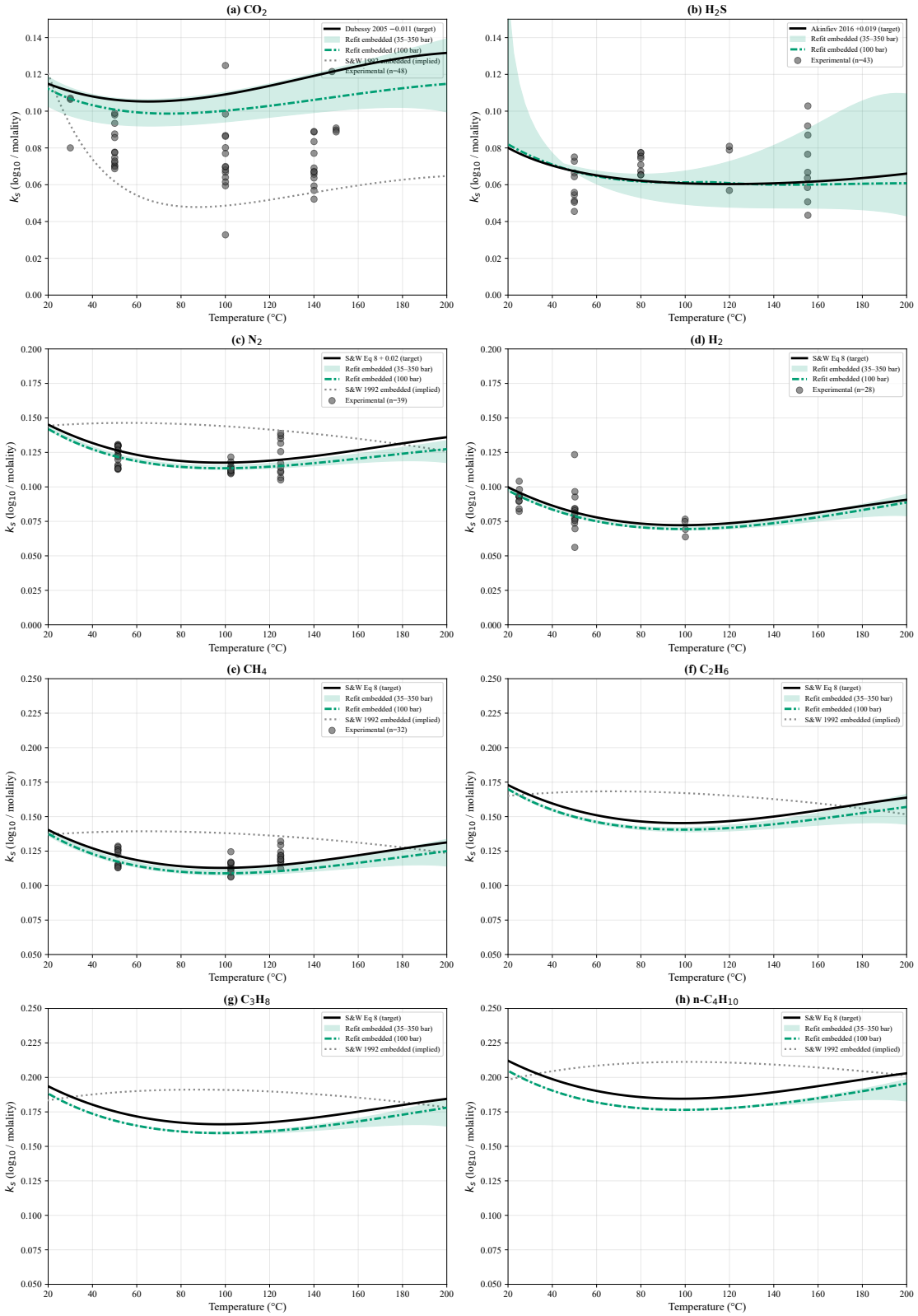


Figure 5: Implied Sechenov coefficient from the refit embedded BIP correlations (green dash-dot lines, $P = 100$ bar) compared against the recommended explicit models (black solid lines; H_2S includes the +0.019 shift) at $m = 1$ mol/kg, all evaluated with Eq. 10 and the brine S&W water alpha. Green shaded bands show the range of implied k_s across 35–350 bar, the artificial pressure sensitivity introduced by embedding salting-out in the BIP. Grey dotted: implied k_s of the S&W 1992 embedded salinity terms with S&W k_{ij}^{NA}. The H_2S band widens below 30°C, where H_2S at the upper pressures is a liquid.

7.4. Validation

Figure 5 compares the implied Sechenov coefficient from embedded BIP correlations against the recommended explicit models.

The embedded correlations reproduce the recommended explicit Sechenov models with mean solubility deviations of 0.6–1.8% for the non-acid gases and 2.9% (CO_2) and 2.4% (H_2S) for the acid gases over each gas's fitting grid (Table 10); the published dry-vapour fit gave 1.56% and 1.99%. The CO_2 and H_2S maxima (15.8% and 19.1%) sit at high temperature at the low- and high-pressure ends of the grid: with the vapour carrying water, a pressure-independent Δk_{ij} follows the change of vapour water content with pressure less closely than it did with a dry vapour. At 100 bar the implied k_s lies up to 0.009 kg/mol below the target for the non-acid gases (gas means 0.002–0.008 kg/mol) (Figure 5), the compromise the fit makes across the pressure grid. These deviations quantify how closely the embedded BIP approach matches the explicit Sechenov correction; actual accuracy relative to experimental brine data depends on the quality of the underlying Sechenov model itself.

Recommendation: Use the embedded BIP approach for most practical applications; it requires no flash routine modification.

8. Discussion

8.1. Implementation of Salinity Effects

The embedded $\Delta k_{ij}(T, m)$ approach (Table 10) is recommended for incorporating salinity effects. It requires no flash routine modification and reproduces the recommended Sechenov models with mean solubility deviations below 3% for all gases and below 2% for the non-acid gases (Section 7.4).

The embedded approach introduces a minor artificial pressure dependence: because the EOS mapping from k_{ij} to dissolved mole fraction x is nonlinear in pressure, the implied Sechenov coefficient varies slightly with pressure even though $\Delta k_{ij}(T, m)$ has no pressure term. Figure 5 illustrates this effect via shaded bands spanning 34–345 bar. For most gases the bands are narrow; CO_2 shows the widest, reflecting stronger EOS nonlinearity near its critical point.

Comparison with original S&W embedding. The original S&W (1992) framework embedded salinity terms (c_s) directly in the BIP correlations for all gases except H_2S (Eqs. 6–8), coupling the freshwater BIP and salting-out in a single expression. The embedded correlations in this work take a different approach: the freshwater BIP $k_{ij,\mathrm{fw}}(T)$ is first fitted independently, and $\Delta k_{ij}(T, m)$ is then fitted explicitly to reproduce the recommended Sechenov model for each gas. This two-stage procedure ensures the embedded form faithfully mimics the character of the underlying k_s model rather than absorbing salting-out effects implicitly during BIP regression.

Alternative implementations were also investigated and are provided in the accompanying code repository [22]. The gamma-phi approach incorporates k_s directly in the flash K-value, $K_i = \gamma_i \phi_i^L / \phi_i^V$. This follows from modelling the aqueous-phase fugacity of each dissolved gas as $f_i^L = x_i \gamma_i \phi_i^L P$, where ϕ_i^L is evaluated from the salt-free EOS with the freshwater BIP and $\gamma_i = 10^{k_s m}$ is the activity coefficient of the dissolved gas referenced to salt-free water at the same temperature and pressure ($\gamma_i \to 1$ as $m \to 0$); the iso-fugacity condition $f_i^L = f_i^V = y_i \phi_i^V P$ then yields the modified K-value. Because the dissolved ions are not components of the EOS, ϕ_i^L carries no salinity dependence and γ_i supplies the measured salt effect. Treating salting-out as an empirical correction to a salt-free model is established practice for gas-brine equilibria, either embedded within the EOS via ion-gas parameters fitted to Sechenov constants [9] or as an activity coefficient on the dissolved gas [4, 10]; the product form used here, an activity coefficient multiplying the EOS fugacity coefficient, was, to our knowledge, first applied to high-pressure gas solubility in brines by Aasberg-Petersen et al. [11], with a Debye-Hückel electrostatic term in place of the Sechenov factor. Explicit Sechenov post-correction of freshwater solubility was also tested. Both alternatives produce solubility predictions equivalent to the embedded Δk_{ij} form, but gamma-phi requires flash routine modification and post-correction introduces material balance errors in multicomponent systems.

K-value initialisation. The standard Wilson K-value correlation produces unphysical $K < 1$ for dissolved gases in gas-water systems, causing flash convergence failures. The accompanying code repository provides fitted K-value initialisation correlations that eliminate these violations for all gases in the S&W framework.

8.2. Comparison with Original S&W

Table 11 summarises the key differences from the original S&W (1992) framework.

Table 11: Comparison of updated framework versus original S&W (1992).

Aspect	S&W (1992)	This Work
Water alpha	Eq. 4 (with embedded salinity)	Eq. 4 retained (Section 2.2.3)
Equilibrium procedure	Not stated in the 1992 paper (Section 3.1)	Coupled, Eq. 10 (Section 3)
k_{ij}^{AQ} fitting	Coupled with salinity terms	Freshwater only + additive $\Delta k_{ij}(T, m)$
Data basis	Few dozen points/gas	~2,000 pointwise values
Salinity approach	Embedded in BIP	Embedded $\Delta k_{ij}(T, m)$ (Table 10)
H_2S salting-out	None (freshwater only)	Via embedded Δk_{ij}, targets Akinfiev [6]
CO_2 salting-out	Embedded in BIP (Eq. 8)	Via embedded Δk_{ij}, targets Dubessy [5]
H_2 support	Not included	Full BIP + Sechenov + embedded Δk_{ij}

8.3. Comparison with Spycher & Pruess (2010) for CO_2

For CO_2, Spycher & Pruess [33] developed a dedicated modified SRK model specifically parameterised for the CO_2-H_2O-NaCl system. A comparison provides context for the accuracy achievable by the general S&W framework versus a single-gas dedicated model.

On freshwater CO_2 solubility data, Spycher & Pruess achieves MARE = 13.0% (611 points) versus 12.4% for this work and 15.3% for the original S&W (606 points, both on the coupled procedure). At low pressures ($P < 50$ bar) Spycher & Pruess gives 18.6% against 17.3% for this work and 24.8% for S&W; at 50–100°C, 5.6% against 7.3% and 10.8%; at $P \geq 500$ bar, 4.5% against 6.1% and 5.4%. The Spycher & Pruess values are those of the published version (611-point basis, five more points at 160–200°C and $P < 50$ bar).

On brine data (109 points, 0.5–5.0 mol/kg), Spycher & Pruess achieves 8.9% versus 17.1% for this work and 18.4% for the original S&W. The dedicated model keeps a clear advantage in brine; in freshwater the general framework now matches it overall.

The S&W framework trades some CO_2 accuracy for the ability to treat eight gases with a consistent methodology and a single flash routine.

8.4. Gas-Gas Binary Interaction Parameters

In multicomponent systems, the mixing rule (Eq. 5) requires BIPs for all component pairs, not just gas-water. The S&W framework focuses on gas-water BIPs (k_{ij}^{AQ} and k_{ij}^{NA}); gas-gas BIPs are adopted from general hydrocarbon VLE literature. These affect multicomponent flash results but have no effect on binary (single-gas + water) calculations.

Table 12 lists the 29 gas-gas BIP values used in this work. All values are temperature-independent constants, standard practice for non-polar and weakly-polar gas-gas interactions where the BIP magnitude is small and the temperature sensitivity negligible compared with gas-water pairs.

The values are conventional constants for these pairs; regressed PR values for most of them are tabulated, per data set, in the compilation of Knapp et al. [20]. As noted above, no regression or validation result in this work is sensitive to them, since all are binary gas–water calculations. All H_2-gas pairs are set to zero due to limited literature on H_2 BIPs with other gases in aqueous-equilibrium contexts. For underground hydrogen storage scenarios where the gas phase is predominantly H_2 with minor impurities, H_2-gas BIP sensitivity is expected to be low. Users with specific multicomponent H_2-rich gas mixtures should validate against experimental VLE data if available. These BIPs are applied identically in both the aqueous and non-aqueous flash calculations. Unspecified pairs (e.g., involving heavier hydrocarbons C_5+) default to zero.

8.5. Henry's Law Consistency

In the low-pressure limit, the PR EOS reduces to Henry's law behaviour. For a single gas dissolved in water at low pressure, the equilibrium condition $f_i^L = f_i^V$ becomes $x_i \gamma_i^* f_i^{0,L} \approx y_i P$, where $\gamma_i^* \to 1$ at infinite dilution in freshwater and $f_i^{0,L}$ is the hypothetical liquid-phase fugacity of the pure gas at system conditions. Rearranging, $x_i \approx P/K_H$ where $K_H = f_i^L / x_i$ is the Henry's constant, equal to $f_i^{0,L}$ in this limit. In the PR EOS framework, K_H is not an explicit input but emerges from the equation of state parameters and the aqueous BIP k_{ij}^{AQ}: at any given temperature, computing the solubility at sufficiently low pressure and dividing P by x yields the implied Henry's constant.

Table 12: Gas-gas binary interaction parameters adopted from literature. All values are constant (temperature-independent).

Gas i	Gas j	k_{ij}	Gas i	Gas j	k_{ij}
CO_2 pairs			*N_2 pairs*		
CO_2	CH_4	0.120	H_2S	N_2	0.170
CO_2	C_2H_6	0.130	C_2H_6	N_2	0.040
CO_2	C_3H_8	0.135	C_3H_8	N_2	0.080
CO_2	nC_4H_{10}	0.130			
CO_2	iC_4H_{10}	0.130	*H_2S pairs*		
CO_2	N_2	−0.020	C_2H_6	H_2S	0.085
CO_2	H_2S	0.097	C_3H_8	H_2S	0.080
CH_4 pairs			*HC–HC pairs*		
CH_4	C_2H_6	0.0026	C_2H_6	C_3H_8	0.001
CH_4	C_3H_8	0.014	C_2H_6	nC_4H_{10}	0.010
CH_4	nC_4H_{10}	0.020	C_3H_8	nC_4H_{10}	0.003
CH_4	iC_4H_{10}	0.020			
CH_4	N_2	0.036	*H_2 pairs (all zero)*		
CH_4	H_2S	0.080	H_2	all gases	0.000

For hydrogen at 25°C, the implied K_H (infinite-dilution aqueous fugacity coefficient times pressure, 1–5 bar) is 7.31–7.33 GPa, 3% above the critically evaluated value of 7.10 GPa [21] (published dry-vapour correlation: 7.20 GPa). The implied $K_H(T)$ reproduces the non-monotonic temperature dependence, with a maximum near 50°C (7.67 GPa at 50°C, 7.58 GPa at 60°C) [21]. For other gases, the implied $K_H(T)$ trends are qualitatively consistent with published Henry's constants.

8.6. Limitations

Key limitations of this work include:

- Limited high-temperature data for some gases (particularly H_2 and N_2)
- Coefficients fitted on Eq. 10; a simulator using the two-flash form incurs the differences of Table 4 (Section 3.3)
- Sechenov models validated primarily for NaCl; extension to mixed brines requires additional work
- Non-aqueous phase BIPs have higher uncertainty than aqueous phase due to limited quality data
- Multicomponent gas-gas interactions use constant literature BIPs (Table 12); co-solubility effects have not been independently validated (Section 8.4)

8.7. Validity Ranges and Error Analysis

The correlations are validated within 273–473 K (gas-dependent; see individual sections), 1–1000 bar (most data below 500 bar), and NaCl molalities of 0–6 mol/kg; the fitted data ranges are summarised in Table A.13 (Appendix Appendix A). Accuracy is expected to degrade outside these ranges, particularly at temperatures approaching the critical point of water where the aqueous and gas-rich phases become increasingly similar. For gases with high aqueous-phase solubility (H_2S and CO_2), the parity plots (Figure A.6a–d) show systematic overprediction at low dissolved-gas fractions and underprediction at high fractions, a consequence of the composition-independent van der Waals one-fluid mixing rule, which applies a single k_{ij} regardless of aqueous-phase composition. This limitation is negligible for sparingly soluble gases.

For CO_2 and H_2S, the fitted $k_{ij}(T)$ represents a best compromise across all compositions at each temperature. In dilute solutions (low x, typically low P) the compromise value is too attractive, while at high dissolved-gas concentrations (high x, typically high P) it is insufficiently attractive. Because pressure and aqueous-phase composition

are strongly correlated for a given gas at fixed temperature, pressure serves as a practical proxy for the composition dependence that the standard mixing rule cannot represent. Analysis of pointwise k_{ij} residuals confirms statistically significant $\partial k_{ij}/\partial(\ln P)$ trends that intensify with temperature, consistent with higher solubility amplifying the composition-dependent mixing limitation. Although pressure-dependent $k_{ij}(T, P)$ forms were investigated, the marginal MARE improvement (a few percentage points for H_2S) does not justify the additional complexity and the requirement for simulators to evaluate k_{ij} as a function of both temperature and pressure within the flash routine. The systematic bias pattern is therefore understood as an inherent limitation of the single-parameter mixing rule.

The elevated mean absolute relative errors reported for H_2S (MARE 31.9% across all 405 freshwater points) are dominated by a small number of low-concentration measurements. Points with $x_{H_2S} < 0.003$ constitute 5% of the dataset (20 points) but contribute 81% of the total absolute error, because the sensitivity $|\partial \ln x/\partial k_{ij}|$ increases sharply as the dissolved-gas fraction decreases, reaching amplification factors of 40–190 at $x < 10^{-3}$ compared with 10–20 at $x \sim 0.01$. At these low concentrations, even modest k_{ij} correlation errors ($\Delta k_{ij} \approx 0.05$) propagate to relative solubility errors exceeding 100%.

Experimental measurement uncertainty is also a contributing factor at low dissolved-gas fractions. The low-x data originate from near-atmospheric experiments ($P \leq 10$ bar) where H_2S partial pressures are low and analytical techniques operate closer to their detection limits. As measurements approach the limit of quantification, reported concentrations are subject to larger relative uncertainty and may exhibit a positive bias due to censoring effects (values below the detection limit are not reported). For the bulk of the dataset at conditions relevant to subsurface applications ($x \geq 0.003$, corresponding roughly to $P > 5$ bar), the MARE is 6.3% (S&W 7.8%, $n = 385$). Eighteen points in the set (Selleck 1952 at 37.8°C, where tabulated x falls with rising pressure; Suleimenov 1994 at or below 60°C) are in error by 250–1,200% for every model; excluding them gives 6.8% (S&W 8.0%, $n = 387$). They are flagged for source checking, not removed.

An additional structural limitation applies near the critical points of CO_2 (304.2 K) and H_2S (373.2 K), where vapour-liquid-liquid equilibrium (VLLE) may occur. The two-phase equilibrium of Eq. 10 does not account for the formation of a second liquid phase rich in CO_2 or H_2S, and the non-aqueous phase is assigned the vapour root of the cubic. At or above the gas's EOS saturation pressure a minimum-Gibbs flash returns the gas-rich liquid instead: for 24 such points in the database (C_3H_8 and H_2S on their saturation lines) an independent multiphase flash differs by −2.6 to +0.4% in x and by a factor of 4–10 in y_{H_2O} [35], while moving no solubility MARE by more than 0.15 pp. At temperatures and pressures near the CO_2 or H_2S saturation curve, especially at high gas partial pressures, the system may split into three phases (aqueous, gas-rich liquid, and vapour). The correlations presented here are not validated for such conditions and should not be applied in VLLE regions without independent phase stability analysis.

9. Conclusions

This work presents updated BIP correlations for the Søreide-Whitson (1992) gas-water-brine framework, based on an expanded experimental database of approximately 2,000 pointwise-regressed binary interaction parameters across eight gases.

1. **Drop-in BIP updates:** New $k_{ij}^{AQ}(T)$ correlations are provided for CO_2, H_2S, CH_4, N_2, H_2, C_2H_6 and C_3H_8 (and He, Appendix Appendix B), fitted using L1-optimal regression on freshwater data only, on the coupled equilibrium procedure. They retain the original S&W water alpha function and replace the original coefficients like for like in existing simulator implementations. Scored on the same procedure, freshwater MARE falls from 15.3% to 12.4% for CO_2, 7.2% to 5.3% for N_2, 11.3% to 8.4% for C_2H_6 and 8.9% to 3.4% for C_3H_8; CH_4 is unchanged (3.9%) and H_2S improves from 7.8% to 6.3% for $x \geq 0.003$.
2. **Embedded salinity corrections:** Additive $\Delta k_{ij}(T, m)$ correlations for all eight gases (and He) reproduce the recommended Sechenov models: Akinfiev et al. [6] for H_2S, Dubessy et al. [5] for CO_2, and Eq. 13 for the non-acid gases (with a +0.02 offset for N_2). Mean solubility deviations from the target models are 0.6–1.8% for the non-acid gases and 2.4–2.9% for H_2S and CO_2.
3. **Hydrogen support:** The framework is extended to hydrogen with rational-form BIP correlations reproducing U-shaped solubility behaviour (MARE = 11.3% across 142 freshwater points), constant $k_{ij}^{NA} = 0.468$ consistent with the S&W volatility trend, and Sechenov coefficient via Eq. 13 ($T_b = 20.3$ K, MAE = 0.0087 kg/mol against 46 experimental points from six sources).

4. **Model sensitivity investigation:** Replacing the S&W alpha function with a higher-accuracy Mathias-Copeman form and adopting a modular gamma-phi salting-out framework both produce indistinguishable solubility predictions when the k_{ij} is refitted (Table 1). The BIP absorbs upstream model differences, validating the simpler approach of retaining the original S&W alpha and embedded salinity formulation.

The updated framework maintains backward compatibility with existing S&W implementations whilst reducing the pointwise k_{ij} MAE for CO_2 from 0.0120 to 0.0087 and for H_2S from 0.0079 to 0.0070, and adding the hydrogen support required for underground storage applications. The gains over S&W are smaller than the published version reported, because that version scored the 1992 coefficients on a dry-vapour basis they were not fitted on.

Data availability

Working Python code for pointwise BIP regression, correlation fitting, and VLE calculations, along with all compiled experimental data, is available at github.com/mwburgoyne/SW_Framework_Refresh. The refit on the coupled procedure (solver, scripts, coefficients and per-point results) is in its `procedure1_refit/` directory (tag `procedure1-2026-11`).

CRediT authorship contribution statement

Mark Burgoyne: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Markus H. Nielsen:** Methodology, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

Roman Symbols

a, b	EOS parameters
a_0, a_1, a_2	Embedded BIP correlation coefficients
f	Fugacity (Pa)
k_{ij}	Binary interaction parameter
K_i	Equilibrium K-value (y_i/x_i)
k_s	Sechenov (salting-out) coefficient (kg/mol)
m	Salt molality (mol/kg H_2O); c_{sw} in S&W notation
P	Pressure (Pa or bar)
R	Universal gas constant (8.314 J/mol·K)
T	Temperature (K or °C as noted)
T_b	Normal boiling point (K)
T_c	Critical temperature (K)
T_r	Reduced temperature (T/T_c)
V	Molar volume (m^3/mol)
x	Mole fraction in aqueous phase
y	Mole fraction in vapour phase
Z	Compressibility factor

Greek Symbols

α	EOS energy parameter temperature function
γ	Activity coefficient
ω	Acentric factor
ϕ	Fugacity coefficient

Subscripts and Superscripts

AQ	Aqueous phase
b	Boiling point
c	Critical property
fw	Freshwater
i, j	Component indices
L	Liquid phase
NA	Non-aqueous phase
r	Reduced property
s	Salting-out (Sechenov)
V	Vapour phase

Abbreviations

BIP	Binary interaction parameter
EOS	Equation of state
MAE	Mean absolute error
MARE	Mean absolute relative error
PR	Peng-Robinson
S&W	Søreide-Whitson
UHS	Underground hydrogen storage
VLE	Vapour-liquid equilibrium

Appendix A. Solubility Parity Plots

Figures A.6 and A.7 present per-gas parity plots comparing predicted versus experimental gas solubility in the aqueous phase, all on the coupled procedure (Eq. 10) and on the evaluation sets of Tables 6 and C.17. Freshwater panels compare the refit correlations with the original S&W correlations (S&W k_{ij}^{NA}). Brine panels compare the refit embedded salinity BIP (recommended), the explicit Sechenov correction applied to the refit freshwater solubility, and the original S&W embedded correlations (Eq. 13 post-factor for H_2S).

Appendix A.1. Fitting Data Extent

Table A.13 summarises the temperature and pressure ranges of the experimental data used to regress the freshwater k_{ij}^{AQ} correlations. The correlations should be applied with caution outside these ranges. In particular, accuracy degrades at temperatures approaching the critical point of water ($T_{c,\mathrm{H_2O}} = 647$ K) where the aqueous and gas-rich phases become increasingly similar and the cubic EOS root selection becomes unreliable.

For gases with high aqueous-phase solubility (H_2S, x up to 0.16; CO_2, x up to 0.14), the parity plots show a systematic pattern: overprediction at low dissolved-gas fractions and underprediction at high fractions. The origin of this pattern in the composition-independent van der Waals one-fluid mixing rule is discussed in Section 8.7; it is negligible for sparingly soluble gases (CH_4, N_2, H_2) that remain in the dilute regime.

Table A.13: Temperature and pressure ranges of the freshwater experimental data used to fit the k_{ij}^{AQ} correlations. n is the number of converged freshwater data points used in each fit. For n-C_4H_{10} no fit was performed (the original S&W form is retained; Section 5.8); its row describes the available data, which extend above 200°C.

Gas	n	T range (°C)	P range (bar)	x_{gas} range
CO_2	606	0–200	0.4–2000	2×10^{-4}–0.087
H_2S	405	21–200	2–309	1×10^{-4}–0.16
CH_4	115	25–171	22–1640	3×10^{-4}–0.010
N_2	127	25–200	4–1013	2×10^{-5}–0.005
H_2	154	0–179	4–1013	4×10^{-5}–0.013
C_2H_6	95	2–200	0.5–3500	3×10^{-5}–0.005
C_3H_8	59	6–149	5–192	8×10^{-5}–7×10^{-4}
n-C_4H_{10}	27	138–238	14–690	7×10^{-5}–0.002

Appendix B. Helium

Helium was added to the framework after the published version and was not part of the peer-reviewed article. It is fitted and scored exactly as the other gases (coupled procedure, Eq. 10; same forms, L1 routine and embedded-Δk_{ij} procedure) and is collected here.

Properties. $T_c = 5.1953$ K, $P_c = 2.28323$ bar, $\omega = -0.38354$, $T_b = 4.2238$ K, from the helium-4 equation of state of Ortiz-Vega et al. as implemented in CoolProp [36].

Data. 296 freshwater points were compiled from five sources; the fit set is 130 points at 20–163°C and 1–1000 bar: Abrosimov and Lebedeva [37] (evaluated values; 105 points, 50–80°C), Gardiner and Smith [38] (12 points, 20–50°C), Gerth [39] (8 points, 25°C), Potter and Clynne [40] (2 points, 104–149°C) and Pray et al. [41] (3 points, 163°C). Brine: 48 points in NaCl, 25–100°C, 51–608 bar, 1.0–4.1 mol/kg (Gardiner and Smith, 33; Gerth, 15).

Aqueous-phase BIP. The rational form gives

$$k_{ij,\mathrm{He}}^{\mathrm{AQ}} = \frac{-71.6155 + T_r}{-0.164407 + 0.5017371\,T_r} \tag{B.1}$$

with $T_r = T/T_c$, $T_c = 5.1953$ K: MAE(k_{ij}) = 0.0184 against the pointwise values and solubility MARE 6.3% ($n = 130$, bias +0.1%). The adopted form is within 0.0004 of the MAE of a five-parameter quartic in T on the same points (Section 5.10). The residuals carry a pressure trend (Figure B.8b): overprediction of solubility at 1–10 MPa and underprediction at 70–100 MPa on Abrosimov and Lebedeva, the structured bias of a $k_{ij}(T)$-only form. The correlation is not validated above 163°C.

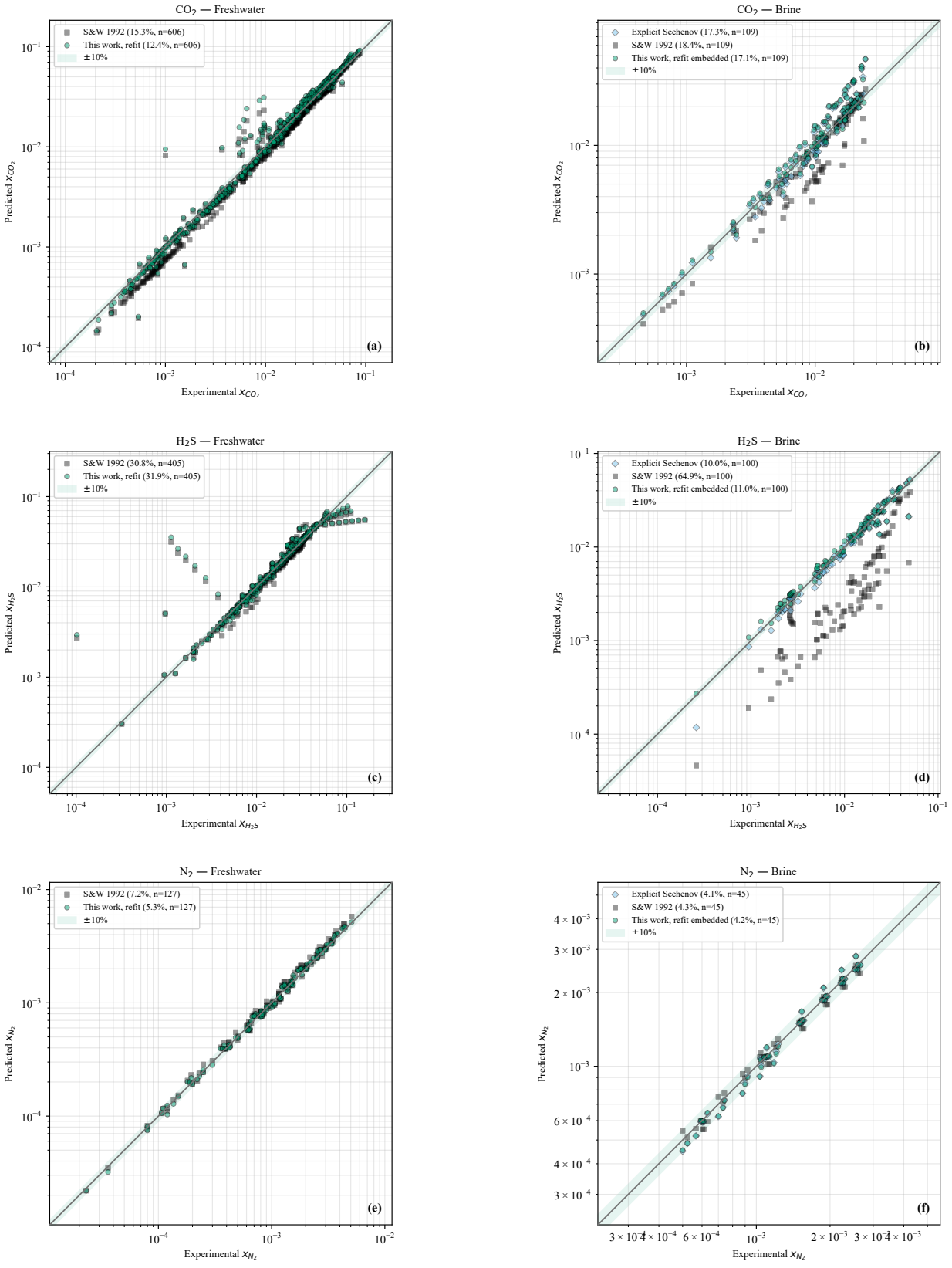


Figure A.6: Solubility parity plots for CO_2, H_2S, and N_2. Left column: freshwater predictions, refit $k_{ij}(T)$ correlations against S&W originals. Right column: brine predictions, explicit Sechenov, refit embedded BIP and S&W original approaches. All on the coupled procedure. MARE values (Eq. 15) shown in legends. Shaded bands indicate ±10%.

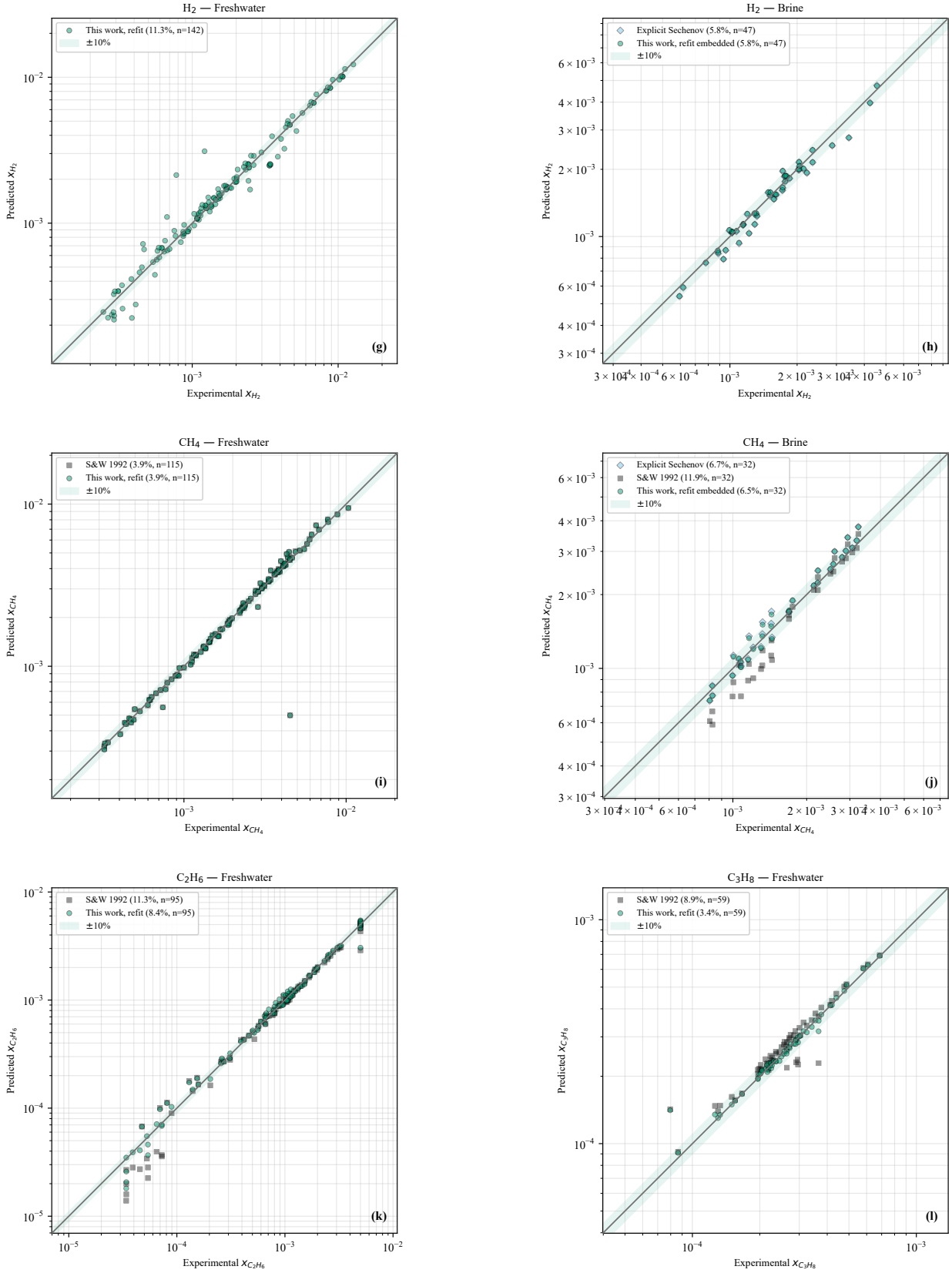


Figure A.7: Solubility parity plots for H_2, CH_4, C_2H_6, and C_3H_8. Layout as in Figure A.6; C_3H_8 has freshwater data only (shown as the final panel), and nC_4H_{10} is omitted as no aqueous data exist. CH_4 brine evaluation uses O'Sullivan & Smith (1970) data only (see Section 6.5).

Table B.14: He freshwater solubility MARE (%) by source on the coupled procedure, bias in brackets.

Source	n	T (°C)	P (bar)	MARE (bias)
Abrosimov & Lebedeva 2013	105	50–80	1–1001	6.8 (+0.2)
Gardiner & Smith 1972	12	20–50	101–608	2.4 (+0.7)
Gerth 1983	8	25	51–405	2.8 (−0.3)
Potter & Clynne 1978	2	104–149	11–13	2.7 (+2.7)
Pray et al. 1952	3	163	14–27	15.9 (−6.2)
All	130	20–163	1–1001	6.3 (+0.1)

Non-aqueous-phase BIP. No He water-content data were found. $k_{ij}^{\mathrm{NA}} = 0.468$, the H_2 value, is used as a surrogate, so the water content of helium-rich gas is unvalidated.

Salting-out. Eq. 13 with $T_b = 4.2238$ K matches k_s extracted from the measured brine/freshwater pairs to a mean absolute deviation of 0.0067 kg/mol ($n = 48$). The embedded correction, fitted to Eq. 13 on the H_2 grid restricted to 25–100°C, is

$$\Delta k_{ij,\mathrm{He}} = (0.363020 - 0.00963936\,T_r + 7.09302 \times 10^{-5}\,T_r^2)\,m \tag{B.2}$$

reproducing the Eq. 13 targets to 0.56% mean and 1.30% maximum solubility deviation. Brine solubility MARE is 12.8% ($n = 48$, bias +3.2%; explicit Sechenov 12.7%), dominated by Gardiner and Smith's 100°C points (+20 to +43%), which disagree with the freshwater measurements of Potter and Clynne [40] at that temperature.

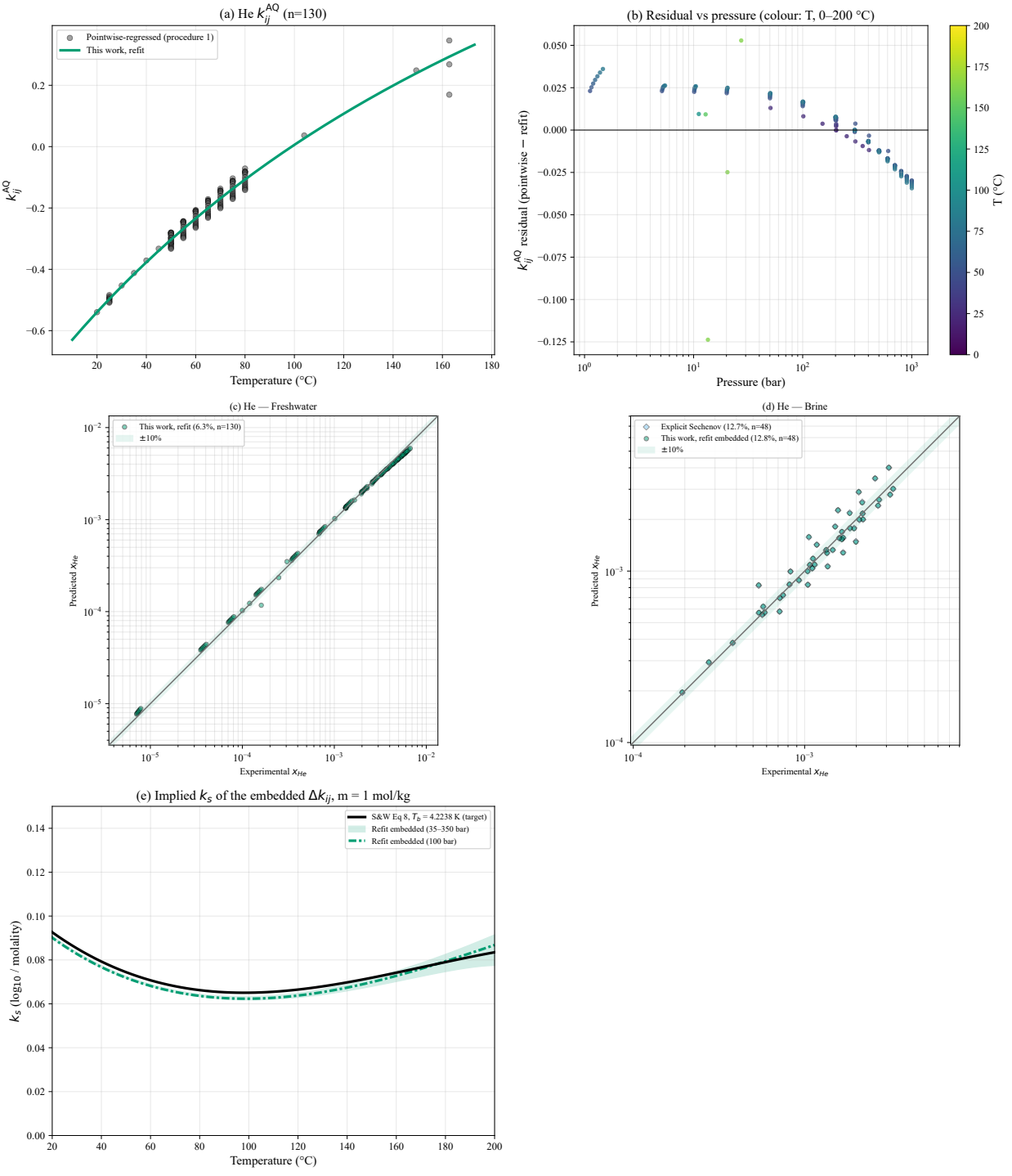


Figure B.8: Helium on the coupled procedure. (a) Pointwise k_{ij}^{AQ} and Eq. B.1. (b) Pointwise residual against pressure, coloured by temperature. (c) Freshwater parity. (d) Brine parity, explicit Sechenov and embedded correction (Eq. B.2). (e) Implied k_s of Eq. B.2 at $m = 1$ mol/kg against the Eq. 13 target; band 35–350 bar.

Appendix C. Changes from the Published Version

The published version [34] regressed every k_{ij}^{AQ} and computed every solubility MARE with the non-aqueous phase treated as pure, dry gas at total pressure. This draft repeats the regression with the coupled equilibrium, Eq. 10 (Section 3). Data, exclusions (plus the five Wang 2021 CO_2 points of Section 4), the 0–200°C window, the functional forms, the L1 routine, the k_{ij}^{NA} grid search and decision rule, the Sechenov targets and grids are unchanged. The changes are: new k_{ij}^{AQ} and Δk_{ij} coefficients; every MARE re-scored on Eq. 10, including the S&W comparison columns; the C_3H_8 and nC_4H_{10} Δk_{ij} fitted on their own freshwater BIPs; helium added (Appendix Appendix B); the comparison against Spycher and Pruess and the H_2 Henry constant re-evaluated. k_{ij}^{NA} constants are unchanged. A section on the choice of equilibrium procedure (Section 3) is new. The MC-3 alpha and gamma-phi comparison (Table 1) was not repeated.

Table C.15 separates the effect of the procedure from that of the refit. In an Eq. 10 calculation the published coefficients underpredict solubility at 150–200°C by 12–29% for CO_2, H_2S, CH_4 and H_2 (Table C.16); the refit removes the bias (−6 to +3%). At or below 60°C the published and refit coefficients agree within 2 pp for every gas except H_2S, where the 18 suspect points dominate: the vapour is nearly dry there and the procedure barely matters.

Table C.15: Freshwater x_{gas} MARE (%), bias (%) in brackets. Pub-P0: published coefficients in the published dry-vapour solve (reproduces the published tables). Pub-P1: published coefficients in an Eq. 10 calculation. SW-P1: S&W 1992 k_{ij}^{AQ} and k_{ij}^{NA} (Eq. 17 for H_2S) on Eq. 10, as they were fitted. New-P1: this draft.

Gas	*n*	Pub-P0	Pub-P1	SW-P1	New-P1
CO_2	611 (606)	16.4 (+10.7)	14.9 (−1.0)	15.3 (−6.3)	**12.4** (+4.8)
H_2S	405	38.5 (+30.4)	38.2 (+13.9)	30.8 (+20.6)	31.9 (+25.7)
H_2S, excl. 18 suspect	387	13.8	13.8	8.0	**6.8**
CH_4	115	4.8 (−0.3)	4.7 (−3.0)	3.9 (−0.6)	**3.9** (−0.7)
N_2	127	5.7 (+1.6)	5.6 (−0.4)	7.2 (+3.7)	**5.3** (+1.2)
H_2	142	**10.2** (+3.0)	12.0 (−1.1)	—	11.3 (+2.3)
C_2H_6	95	12.9 (+4.6)	9.2 (−2.4)	11.3 (−5.5)	**8.4** (+0.8)
C_3H_8	59	5.0 (+1.4)	4.8 (−1.2)	8.9 (+4.8)	**3.4** (+1.2)

Table C.16: As Table C.15, 150–200°C only.

Gas	*n*	Pub-P0	Pub-P1	SW-P1	New-P1
CO_2	113	30.0 (+23.3)	20.7 (−20.4)	8.4 (−6.0)	9.0 (−2.4)
H_2S	103	31.9 (+20.7)	28.8 (−28.8)	8.1 (+2.2)	6.8 (+2.9)
CH_4	21	12.7 (−2.9)	12.8 (−12.2)	9.3 (−5.3)	9.2 (−5.9)
N_2	11	6.9 (+3.5)	5.8 (−5.8)	5.3 (−5.3)	3.3 (−0.8)
H_2	20	9.1 (+3.4)	17.2 (−13.3)	—	11.6 (−2.9)
C_2H_6	26	14.1 (+8.5)	6.9 (−5.2)	6.0 (−3.2)	5.8 (+0.9)

Table C.17: Brine x_{gas} MARE (%), bias (%) in brackets; columns as Table C.15, embedded salinity corrections throughout (S&W: Eqs. 6–8, Eq. 13 post-factor for H_2S).

Gas	*n*	Pub-P0	Pub-P1	SW-P1	New-P1
CO_2	109	17.3 (+10.0)	18.0 (+7.3)	18.4 (−15.2)	17.1 (+11.6)
H_2S	100	15.4 (+8.1)	7.6 (−2.7)	64.9 (−64.9)	11.0 (+4.3)
CH_4	32	5.6 (+2.4)	5.1 (+1.7)	11.9 (−9.8)	6.5 (+3.3)
N_2	45	4.6 (−2.8)	4.8 (−3.3)	4.3 (−1.3)	4.2 (−1.5)
H_2	47	4.1 (−0.3)	3.9 (−0.8)	—	5.8 (−2.5)

In brine the published coefficients score better than the refit on H_2S, CH_4 and H_2 in an Eq. 10 calculation (Table C.17). Splitting each prediction into its freshwater baseline and its salt factor shows why: the mean salt factors

of the two coefficient sets agree within 0.4–2.9% per gas, and the difference comes from the freshwater baseline. For Suleimenov (1994) H_2S at 155°C the published baseline is 28% low on Eq. 10, which offsets a salt correction that is too weak at that temperature (brine error −9.7%); with the corrected baseline the refit salt correction gives +17.1%. The refit exposes a high-temperature weakness in the H_2S Sechenov model rather than introducing an error.

Figure C.9 shows the size of the change. Paired at the same measured point, the pointwise k_{ij}^{AQ} on Eq. 10 is lower than the dry-vapour value at all 1,561 points, by less than 0.001 at or below 60°C and by 0.013 (C_2H_6) to 0.086 (H_2) as a gas mean at 150–200°C, several times the pointwise scatter. Measurement noise cancels in this paired difference.

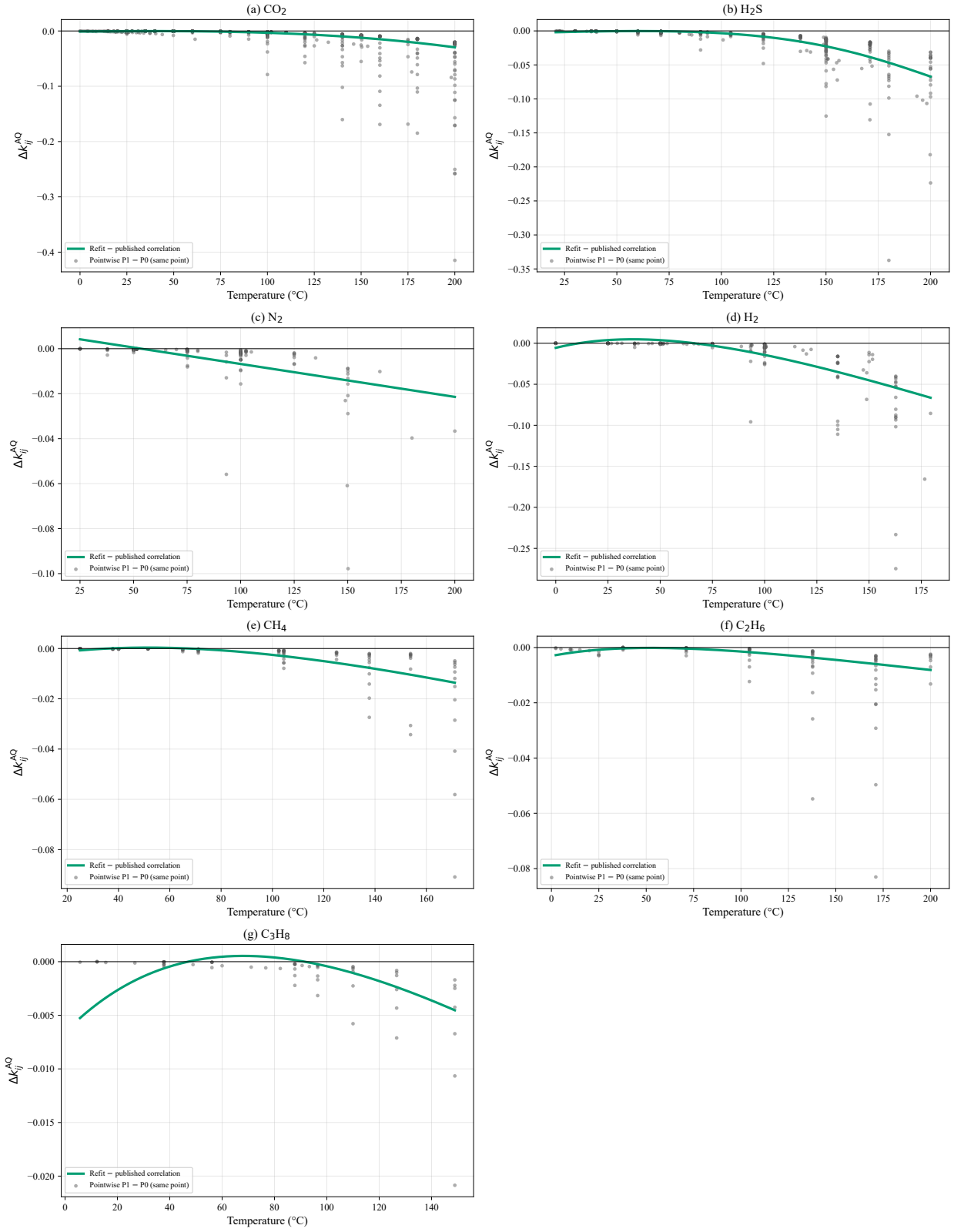


Figure C.9: Refit minus published k_{ij}^{AQ} correlation (green) and the paired pointwise shift (grey: pointwise value on Eq. 10 minus the dry-vapour value at the same measured point).

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